

Measuring Financial Inclusion Across Indian States: A Two-Stage PCA Approach



RESERVE BANK OF INDIA

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Summer Placement Scheme – 2026

DECLARATION

I hereby declare that the project report titled “**Measuring Financial Inclusion Across Indian States: A Two-Stage PCA Approach**” has been completed by me during the period June 2026 to July 2026 as a part of the Summer Placement Scheme (SPS – 2026) of the Reserve Bank of India, Kolkata.

I further declare that the report is the result of my own efforts and has not been copied from any other source. The data used in this report is from publicly available databases and government sources. The views expressed are personal and do not represent the views of the Reserve Bank of India.

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CERTIFICATE OF APPROVAL

This is to certify that **Anwesh Das**, a student of MA Economics (Specialisation in Trade and Finance) at Indian Institute of Foreign Trade (IIFT), Kolkata, has successfully completed the internship project titled “**Measuring Financial Inclusion Across Indian States: A Two-Stage PCA Approach**” under my guidance and supervision at the Department of Statistics and Information Management (DSIM), Reserve Bank of India, Kolkata from 1st June 2026 to 31st July 2026.

This project was carried out as a part of the Summer Placement Scheme – 2026 and fulfills the requirements laid down by the Reserve Bank of India for internship submission.

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1. INTRODUCTION

1.1 Background and Motivation

Financial inclusion has become a key component of inclusive economic development and an important policy objective for governments and financial regulators worldwide. Access to formal financial services enables individuals and businesses to save, obtain credit, manage financial risks, make payments and invest in productive activities. Greater participation in the formal financial system improves household welfare, supports entrepreneurship and contributes to sustainable economic growth, whereas financial exclusion limits economic opportunities, reinforces income inequalities and increases dependence on costly informal finance (Honohan, 2008; Sarma & Pais, 2011). Consequently, expanding access to affordable and appropriate financial services has become central to strategies for inclusive and equitable development

The concept of financial inclusion has evolved from a narrow emphasis on expanding banking access to a broader focus on the availability, accessibility and effective usage of formal financial services. Physical access alone does not ensure meaningful financial inclusion, as individuals may own bank accounts without actively using formal financial services because of low financial literacy, limited income, unsuitable financial products or high transaction costs. This broader perspective is reflected in the Global Findex 2024, which reported that global account ownership increased from 51 percent in 2011 to 79 percent in 2024 (World Bank, 2025).

In India, financial inclusion has been a longstanding public policy priority. Major initiatives such as bank nationalisation, the Lead Bank Scheme, Priority Sector Lending and the Self Help Group-Bank Linkage Programme expanded the reach of formal finance, particularly in rural and underserved areas. The Business Correspondent model, introduced in 2006, further strengthened outreach by delivering banking services through authorised agents where establishing branches was not economically viable.

Financial inclusion accelerated after the launch of the Pradhan Mantri Jan Dhan Yojana (PMJDY) in 2014. Together with the JAM (Jan Dhan, Aadhaar, Mobile) Trinity, Digital India and the rapid expansion of the Unified Payments Interface (UPI), it transformed India's financial ecosystem by expanding banking access and promoting digital financial services. By February 2025, PMJDY had opened over 54.97 crore accounts with deposits exceeding ₹2.52 lakh crore. To monitor progress more comprehensively, the Reserve Bank of India introduced the multidimensional Financial Inclusion Index, which increased from 60.1 in March 2023 to 64.2 in March 2024 (Reserve Bank of India, 2024).

Despite this progress, substantial interstate disparities persist. These differences suggest that no single indicator can adequately capture the level of financial inclusion. A multidimensional composite measure is therefore required to provide a comprehensive assessment across states.

1.2 Need for a Composite Financial Inclusion Index

Financial inclusion is inherently multidimensional and cannot be adequately measured using a single indicator. Traditional measures such as bank branches, bank accounts, deposits, credit outstanding or automated teller machines each capture only one aspect of financial inclusion. Although these indicators provide valuable information individually, none reflects the overall extent of financial inclusion (Sarma, 2008; OECD, 2008).

Recognising these limitations, Sarma (2008) conceptualised financial inclusion through three interrelated dimensions: banking penetration, availability of financial services and usage of financial services. States may perform differently across these dimensions. A state with an extensive banking network may still experience low usage of formal financial services, while another may achieve high account ownership and credit utilisation despite relatively limited physical infrastructure. Assessing financial inclusion using a single indicator can therefore produce incomplete or misleading conclusions. Composite indices overcome this limitation by integrating multiple indicators into a single summary measure. The OECD Handbook on Constructing Composite Indicators identifies composite indices as appropriate for measuring complex multidimensional phenomena, enabling meaningful comparisons across regions and over time (OECD, 2008).

Several methodologies have been proposed for constructing composite financial inclusion indices, differing in their choice of indicators, weighting schemes and aggregation procedures. Among these, Principal Component Analysis (PCA) is widely used because it derives objective statistical weights from the data while accounting for correlations among indicators (OECD, 2008). More recent studies have adopted hierarchical or two-stage PCA approaches, which estimate dimension-specific scores before constructing the overall index, thereby preserving the multidimensional structure of financial inclusion (Cámara & Tuesta, 2014; Kasturi & Annemalla, 2024). Sarma's distance-based methodology follows the same principle by first constructing dimension scores and then combining them into a composite measure. To ensure conceptual comparability while retaining the statistical advantages of PCA, this study adopts a hierarchical two-stage PCA framework.

1.3 Research Problem

Despite substantial improvements in financial inclusion, considerable interstate disparities persist in India. While the Reserve Bank of India's Financial Inclusion Index (FI-Index) provides an official multidimensional measure at the national level, it does not produce comparable estimates for individual states. Consequently, there is no official benchmark for systematically assessing and comparing financial inclusion across states over time.

This study addresses this gap by developing a hierarchical two-stage Principal Component Analysis (PCA)-based Financial Inclusion Index for 28 Indian states, validating the methodology against the RBI FI-Index at the national level, and comparing the resulting state-level estimates with Sarma's distance-based methodology.

1.4 Research Objectives

The primary objective of this study is to construct and compare state-level Financial Inclusion Indices (FII) for India using a two-stage Principal Component Analysis (PCA) for the period 2016–17 to 2024–25, and to examine interstate disparities in financial inclusion. To achieve this objective, the study pursues the following specific objectives:

1. To construct a Financial Inclusion Index (FII) using a two-stage Principal Component Analysis (PCA), validate the national-level index against the Reserve Bank of India's Financial Inclusion Index (RBI FI-Index), and apply to estimate comparable FIIs for 28 Indian states from 2016–17 to 2024–25.
2. To construct state-level FIIs using Sarma's distance-based methodology and compare the FIIs obtained from the two-stage PCA and Sarma approaches in order to assess their consistency and robustness.
3. To analyse the spatial and temporal evolution of financial inclusion across Indian states by examining interstate disparities, regional patterns, convergence dynamics and changes in relative performance over the study period.

1.5 Scope and Significance of the Study

This study examines financial inclusion across 28 Indian states during 2016–17 to 2024–25. It measures financial inclusion using multidimensional indices, compares two established methodologies, and analyses interstate disparities and their evolution over time.

The study is significant from both academic and policy perspectives. Academically, it demonstrates PCA and Sarma approach can be constructed on a common three-dimension framework and directly compared. The PCA-based index is also validated at the national level against the RBI FI-Index, an approach that has rarely been undertaken in the Indian financial inclusion literature. From a policy perspective, the state-level analysis identifies interstate and regional disparities and highlights the dimensions of financial inclusion requiring greater policy attention, thereby supporting more targeted and evidence-based interventions.

2. LITERATURE REVIEW

2.1 Concept and Meaning of Financial Inclusion

While Chapter 1 introduced financial inclusion as a policy priority, the literature provides several definitions, with no single definition universally accepted. Broadly, financial inclusion refers to ensuring that individuals and businesses have access to affordable, appropriate and timely financial products and services through the formal financial system (Honohan, 2008). Earlier definitions focused primarily on banking access, whereas more recent perspectives also emphasise the availability and effective usage of financial services.

The concept is closely related to financial exclusion, which concerns the barriers preventing individuals and groups from participating in the formal financial system. Reflecting this broader perspective, the Rangarajan Committee (2008) extended the concept beyond bank account ownership to include savings, credit, insurance, payment and remittance services, a definition that continues to influence India's financial inclusion policy.

A major contribution to the literature was made by Sarma (2008), who conceptualised financial inclusion as a multidimensional phenomenon comprising banking penetration, availability and usage. This framework has become one of the most widely adopted conceptual models in financial inclusion research and provides the theoretical basis for the methodologies employed in this study.

The literature consistently associates greater financial inclusion with improved savings, access to institutional credit, investment and broader economic participation (Honohan, 2008; Sarma & Pais, 2011). This multidimensional understanding has encouraged researchers to develop composite indices capable of measuring financial inclusion more comprehensively. The next section reviews the evolution of financial inclusion before examining these measurement approaches.

2.2 Evolution of Financial Inclusion

Financial inclusion has evolved from a narrow emphasis on expanding banking access to a broader objective of ensuring that individuals and businesses can access and effectively use affordable, appropriate and quality financial services. This shift reflects the growing recognition of financial inclusion as a key driver of inclusive economic development, poverty reduction and social welfare (Honohan, 2008; World Bank, 2025).

Globally, financial inclusion gained prominence during the early 2000s as international organisations such as the World Bank, the United Nations and the G20 promoted it as a policy priority for inclusive growth (Gupte et al., 2012). The launch of the Global Findex Database in 2011 further strengthened global monitoring by providing internationally comparable measures of financial access and usage (World Bank, 2025).

In India, the focus similarly shifted from expanding banking infrastructure to promoting meaningful access and usage of financial services. This multidimensional perspective was reflected in the Reserve Bank of India's Financial Inclusion Index (FI-Index), introduced in 2021, which measures financial inclusion through the dimensions of access, usage and quality (Reserve Bank of India, 2024). The National Strategy for Financial Inclusion (2025–2030) further reinforces this approach by emphasising digital financial services, financial literacy and consumer protection.

Despite sustained national progress, substantial interstate disparities persist, indicating that national averages conceal important regional variation. These differences have encouraged researchers to develop multidimensional state-level measures of financial inclusion, which are reviewed in the following section.

2.3 Measurement Approaches to Financial Inclusion

As financial inclusion is inherently multidimensional, researchers have developed a range of measurement approaches, from simple banking outreach indicators to composite indices. These approaches have evolved alongside a broader understanding of financial inclusion, recognising that meaningful inclusion depends on the availability and effective usage of financial services in addition to access (Sarma, 2008; OECD, 2008).

2.3.1 Single-Indicator Approaches

Early studies measured financial inclusion using individual indicators such as bank branches, ATMs, deposit accounts and credit accounts. Although these indicators were simple and readily available,

each captured only one aspect of financial inclusion, often providing an incomplete assessment of the overall concept (Sarma, 2008; Gupte et al., 2012).

2.3.2 Distance-Based Composite Indices

To overcome these limitations, researchers developed composite indices that combine multiple dimensions of financial inclusion. Sarma (2008) introduced the Index of Financial Inclusion (IFI), which measures banking penetration, availability and usage using a normalised inverse Euclidean distance approach. The methodology was subsequently refined while retaining the same conceptual framework (Sarma, 2012), establishing one of the earliest and most influential multidimensional approaches to financial inclusion measurement.

2.3.3 PCA-Based Composite Indices

Principal Component Analysis (PCA) has become one of the most widely used methods for constructing financial inclusion indices because it derives objective weights from the correlation structure of the data rather than assigning equal or predetermined weights (OECD, 2008; Nicoletti et al., 2000) and Broby and Smyth (2025) argue that this data-driven approach improves transparency and reduces subjectivity in index construction.

More recent studies have extended PCA through hierarchical or two-stage frameworks that first estimate dimension-specific scores before combining them into an overall index. This preserves the multidimensional structure of financial inclusion while retaining the advantages of statistically derived weights. Building on the framework proposed by Cámara and Tuesta (2014), Kasturi and Annemalla (2024) applied a two-stage PCA approach to measure financial inclusion across Indian states. Other studies have further expanded financial inclusion measurement using factor analysis and indicators of digital financial services (Amidžić et al., 2014; Gharbi & Kammoun, 2023).

Despite these advances, no single methodology has gained universal acceptance. Existing approaches continue to differ in their weighting, normalisation and aggregation procedures, a methodological diversity reviewed further in the following section.

2.4 Previous Studies on Financial Inclusion

The growing importance of financial inclusion has generated an extensive body of literature examining its measurement, determinants and developmental implications. As the concept evolved from banking access to a multidimensional phenomenon, researchers proposed a variety of composite indices that differ in their choice of indicators, weighting methods and aggregation

procedures. The literature may broadly be classified into international studies, national studies on India and state-level studies.

2.4.1 International Studies

International research established financial inclusion as an important driver of inclusive economic development and simultaneously advanced its measurement. Honohan (2008) demonstrated a strong association between financial access, poverty reduction and economic development. Subsequent studies shifted towards multidimensional measurement, with Cámara and Tuesta (2014) proposing a two-stage Principal Component Analysis (PCA) framework, Amidžić et al. (2014) developing a composite index for cross-country comparisons, and Gharbi and Kammoun (2023) refining financial inclusion measurement through additional dimensions and improved weighting procedures. Collectively, these studies illustrate the transition from simple banking indicators to more comprehensive multidimensional approaches.

2.4.2 Studies on Financial Inclusion in India

Indian research similarly progressed from measuring banking outreach to constructing multidimensional financial inclusion indices. Gupte et al. (2012) argued that financial inclusion extends beyond banking access and requires composite measures capturing multiple dimensions of the formal financial system. Subsequent studies expanded this framework by incorporating supply-side, demand-side and institutional factors (Ambarkhane et al., 2016), developing multidimensional composite indices (Goel & Sharma, 2017), and demonstrating the importance of transparent weighting procedures in financial inclusion measurement (Kapoor, 2023). Overall, these studies strengthened financial inclusion measurement in India while highlighting that differences in indicators and weighting schemes can influence index estimates.

2.4.3 State-Level Studies in India

Recognising India's substantial regional disparities, several studies examined financial inclusion at the state level. Dixit and Ghosh (2013) emphasised the importance of interstate analysis for understanding inclusive growth. Yadav et al. (2021) constructed a multidimensional index for Indian states and reported persistent regional disparities despite improvements in account ownership following the Pradhan Mantri Jan Dhan Yojana (PMJDY). Barik et al. (2022) demonstrated a positive association between financial inclusion and human development using PCA and panel regression. Kasturi and Annemalla (2024) applied a hierarchical two-stage PCA framework and documented substantial interstate differences in financial inclusion, while Sahani and Sharmacharjee (2026) employed Sarma's distance-based methodology for the North-Eastern states, illustrating the

continued relevance of alternative multidimensional approaches. Saravanabhavan (2018) further broadened financial inclusion measurement by incorporating cooperative banking institutions and factor analysis.

Taken together, the literature has substantially advanced the measurement of financial inclusion at both national and state levels. However, differences in indicator selection, weighting methods, normalisation procedures and aggregation techniques continue to produce varying estimates and state rankings, highlighting the research gaps discussed in the following section.

2.5 Research Gap

Existing studies rarely compare alternative methodologies using the same dataset, variables and time period, making it difficult to assess how methodological choices influence financial inclusion estimates and state rankings. Although two-stage PCA has recently been introduced into the financial inclusion literature, its application to state-level measurement in India remains limited. In addition, most Indian studies do not validate newly constructed indices against the official Reserve Bank of India Financial Inclusion Index (RBI FI-Index), despite its value as a benchmark for assessing reliability.

Relatively few studies provide long-term, comparative assessments while simultaneously examining how differences in weighting, normalisation and aggregation affect the measurement of financial inclusion.

To address these gaps, the present study develops a two-stage Principal Component Analysis (PCA)-based Financial Inclusion Index and validates it against the official RBI Financial Inclusion Index (RBI FI-Index) at the national level. The validated methodology is then applied to construct comparable indices for 28 Indian states during 2016–17 to 2024–25. Using the same dataset, the study also constructs state-level indices based on Sarma's distance-based methodology, enabling a systematic comparison of the two approaches while providing a comprehensive assessment of interstate financial inclusion in India.

3. DATA AND VARIABLES

3.1 Data Sources and Coverage

This study constructs a balanced panel dataset for 28 Indian states covering 2016–17 to 2024–25. The dataset comprises indicators representing the penetration, availability and usage dimensions of

financial inclusion, compiled from official government publications, statutory institutions and recognised industry sources.

A national annual dataset covering 2004–05 to 2024–25 was also compiled from the same sources. This dataset was used to estimate the hierarchical two-stage Principal Component Analysis (PCA) model, derive the dimension and composite loadings applied to the state-level Financial Inclusion Index, and validate the resulting national index against the Reserve Bank of India's Financial Inclusion Index (RBI FI-Index).

The RBI FI-Index is constructed using a comprehensive framework comprising 97 indicators spanning the dimensions of Access, Usage and Quality (Reserve Bank of India, 2024). Replicating this framework at the state level is not feasible because many constituent indicators are unavailable on a consistent state-wise basis or are not reported over a sufficiently long time horizon. Consequently, this study selected a set of ten indicators that are consistently available for both the national and state-level datasets throughout the study period. These indicators provide broad coverage of the penetration, availability and usage dimensions while ensuring methodological consistency and comparability across states and over time.

Table 3.1 Data Sources

VARIABLE	SOURCE
Deposits	RBI Handbook of Statistics on Indian States
Credit Outstanding	RBI Basic Statistical Returns (BSR-1)
Insurance Premium	IRDAI Handbook on Indian Insurance Statistics
Number of Offices of Scheduled Commercial Banks	RBI Handbook of Statistics on Indian States
Deposit Accounts	EPWRF Database
Rural Offices of Scheduled Commercial Banks	EPWRF Database
Bank Employees	EPWRF Database
Insurance Offices	IRDAI Handbook on Indian Insurance Statistics
Internet Subscribers	Indian Telecom Services Performance Indicators Report
Credit Accounts	RBI Basic Statistical Returns (BSR-1)
Population	UIDAI State/UT-wise Aadhaar Saturation Tables

Note: The FY 2023–24 insurance premium values for Manipur, Meghalaya, Nagaland, Sikkim and Tripura exhibited anomalous spikes and were replaced by linear interpolation using the adjacent years (2022–23 and 2024–25) before calculating Insurance Density. State-wise internet subscriber data for 2016–17 to 2020–21 were unavailable for certain states because the Indian Telecom Services Performance Indicators Report reported combined state totals. These were apportioned according to population shares. Population data were obtained from the UIDAI State/UT-wise Aadhaar Saturation Tables because the 2011 Census is outdated and the 2021 Census has not yet been conducted.

Population data were obtained from the UIDAI State/UT-wise Aadhaar Saturation Tables, as the 2011 Census is outdated and the 2021 Census has not yet been conducted.

The raw variables were transformed into per capita, per lakh population and ratio-based indicators to ensure interstate comparability. These derived variables constitute the final dataset used for index construction.

3.2 Variables and Justification

Consistent with the three-dimensional framework adopted in this study (Sarma, 2008), ten derived indicators representing penetration, availability and usage were used to construct the Financial Inclusion Index. Table 3.2 presents the derived variables, while Table 3.3 summarises their justification.

Table 3.2 Derived Variables for Financial Inclusion Index Construction

Code	Derived Variable	Unit
DEP_PC	Deposits per capita	₹ per person
CR_OUT_PC	Credit Outstanding per capita	₹ per person
INS_DEN	Insurance Density (Insurance Premium per capita)	₹ per person
SCB_OFF	Scheduled Commercial Bank Offices	Per lakh population
DEP_ACC	Deposit Accounts	Per lakh population
SCB_OFF_R	Rural Scheduled Commercial Bank Offices	Per lakh population
EMP	Bank Employees	Per lakh Deposit Accounts
INS_OFF	Insurance Offices	Per lakh population
INT_SUB	Internet Subscribers	Per lakh population
CR_ACC	Credit Accounts	Per lakh population

Table 3.3 Variable Justification

Variable	Formal Definition	Dimension	Justification for Inclusion
DEP_ACC	Number of deposit accounts per lakh population.	Penetration	Measures the extent to which the population is connected to the formal banking system through deposit account ownership. It is one of the most widely used indicators of banking penetration.
CR_ACC	Number of credit accounts per lakh population.	Penetration	Captures the breadth of formal credit inclusion by reflecting the proportion of the population with access to institutional credit.
SCB_OFF	Number of scheduled commercial bank offices per lakh population.	Availability	Measures the physical availability of formal banking infrastructure and is a core indicator of banking outreach and accessibility.
SCB_OFF_R	Number of rural scheduled commercial bank offices per lakh population.	Availability	Captures the availability of banking infrastructure in rural areas, where financial exclusion is generally more pronounced, thereby reflecting rural financial outreach.
EMP	Number of scheduled commercial bank employees per lakh Deposit Accounts.	Availability	Reflects the operational capacity of the banking system to deliver financial services. As this variable has received limited attention in previous financial inclusion studies, it is included in the present study as an exploratory indicator of banking service capacity.
INS_OFF	Number of insurance offices per lakh population.	Availability	Measures the physical accessibility of insurance services and recognises insurance as an important component of multidimensional financial inclusion.
INT_SUB	Number of internet subscribers per lakh population.	Availability	Serves as a proxy for the digital infrastructure required to access internet-based financial services, thereby supporting digital financial inclusion.
DEP_PC	Total deposits held in scheduled commercial banks divided by the state population (₹ per capita).	Usage	Reflects the utilisation of formal banking services for savings and the depth of financial participation.
CR_OUT_PC	Total outstanding bank credit divided by the state population (₹ per capita).	Usage	Measures the intensity of formal credit utilisation and complements credit account ownership by capturing the volume of institutional credit accessed.
INS_DEN	Total insurance premium divided by the state population	Usage	Reflects the utilisation of insurance services and the extent of financial risk protection through formal insurance coverage.

4. PCA METHODOLOGY

4.1 Justification for Using Principal Component Analysis

Constructing a Financial Inclusion Index (FII) requires combining multiple indicators into a single composite measure. This study employs ten indicators representing the penetration, availability and usage dimensions of financial inclusion. Because these indicators are conceptually related and correlated, an appropriate weighting method is essential (OECD, 2008). Equal and judgement-based weighting schemes either assume identical importance across indicators or rely on subjective judgement. The OECD therefore recommends data-driven weighting methods for correlated indicators (OECD, 2008; Amidžić et al., 2014; Cámara & Tuesta, 2014).

Principal Component Analysis (PCA) derives objective weights from the correlation structure of the data by transforming correlated variables into a smaller set of uncorrelated components. The first principal component (PC1), which captures the greatest proportion of variation, provides the weights for index construction while reducing redundancy among correlated indicators. Consequently, PCA has become one of the most widely used methods for constructing composite indices generally (Broby & Smyth, 2025) and financial inclusion indices in particular (Cámara & Tuesta, 2014; Amidžić et al., 2014; Tram et al., 2023).

Rather than applying PCA directly to all ten indicators, this study adopts a hierarchical two-stage framework based on Sarma's three-dimensional framework of penetration, availability and usage (Sarma, 2008; Sarma, 2012). Separate PCAs are first estimated for each dimension to obtain latent dimension scores, which are subsequently combined through a second-stage PCA to construct the composite Financial Inclusion Index. This approach preserves the multidimensional structure of financial inclusion while retaining the advantages of statistically derived weights, consistent with the frameworks proposed by Cámara and Tuesta (2014) and Kasturi and Annemalla (2024). The following sections assess data suitability, describe the estimation procedure and present the construction and validation of the index.

4.2 Assessment of Data Suitability for Principal Component Analysis

Before applying the two-stage PCA, the dataset was evaluated using the correlation matrix, the determinant of the correlation matrix, the Kaiser-Meyer-Olkin (KMO) measure of sampling adequacy and Bartlett's Test of Sphericity. These diagnostics assess whether the correlation structure is appropriate for principal component extraction (OECD, 2008).

The correlation matrix (Figure 4.1) shows strong relationships among the selected indicators, indicating substantial common variation. Most indicators are positively correlated, whereas the employee variable (EMP) exhibits strong negative correlations with the remaining indicators, an inverse relationship examined later when interpreting the Stage 1 component loadings. The determinant of the correlation matrix is extremely small (2.09×10^{-17}), indicating strong intercorrelation and supporting the application of PCA.

The diagnostic statistics in Table 4.1 further confirm the suitability of the dataset. The overall KMO statistic of 0.78 exceeds the recommended threshold of 0.60, indicating adequate sampling adequacy. Bartlett's Test of Sphericity is highly significant ($\chi^2 = 608.14$, $p < 0.001$), rejecting the null hypothesis that the correlation matrix is an identity matrix. Together, these results demonstrate that the indicators share sufficient common variation to justify the hierarchical PCA framework adopted in this study.

Table 4.1 Assessment of Data Suitability for PCA

Diagnostic Test	Result	Interpretation
Determinant of Correlation Matrix	2.09×10^{-17}	Indicates strong intercorrelation among indicators
Overall KMO Statistic	0.78	Adequate sampling adequacy for PCA
Bartlett's Test (χ^2)	608.14	Significant
Degrees of Freedom	45	—
<i>p</i> -value	< 0.001	Rejects the null hypothesis of an identity correlation matrix

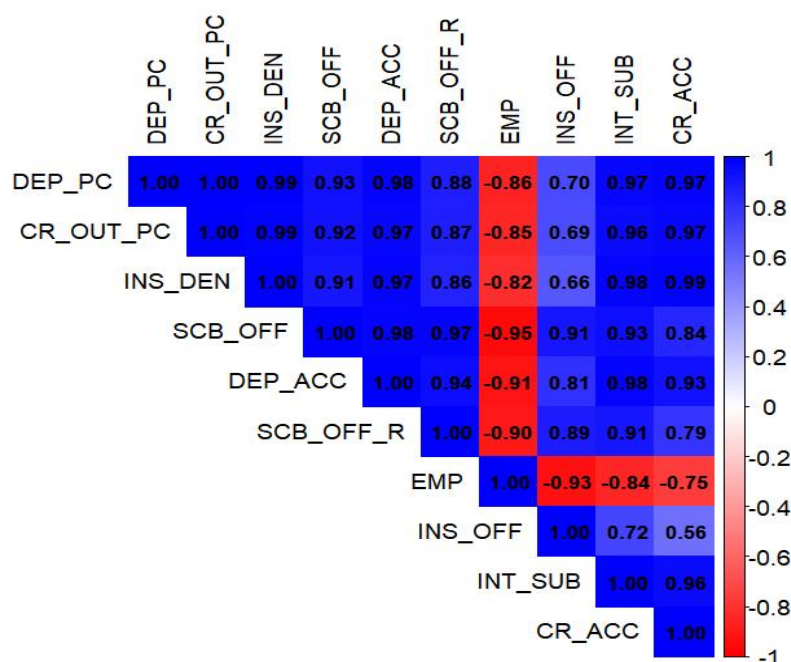


Figure 4.1 Correlation Matrix of Financial Inclusion Indicators

4.3 Hierarchical Two-Stage Principal Component Analysis

Following the satisfactory PCA diagnostic results, the hierarchical two-stage PCA framework described in Section 4.1 was implemented to construct the Financial Inclusion Index. In Stage 1, separate principal component analyses were performed for the penetration, availability and usage dimensions. In Stage 2, the resulting dimension scores were combined to obtain the composite Financial Inclusion Index.

4.3.1 Classification of Financial Inclusion Indicators

Consistent with the multidimensional framework proposed by Sarma (2008, 2012), the ten financial inclusion indicators were classified into three dimensions: penetration, availability and usage.

- **Penetration:** Deposit Accounts per lakh Population (DEP_ACC) and Credit Accounts per lakh Population (CR_ACC).
- **Availability:** Scheduled Commercial Bank Offices per lakh Population (SCB_OFF), Rural Scheduled Commercial Bank Offices per lakh Population (SCB_OFF_R), Employees per lakh Deposit Accounts (EMP), Insurance Offices per lakh Population (INS_OFF), and Internet Subscribers per lakh Population (INT_SUB).
- **Usage:** Deposits per Capita (DEP_PC), Credit Outstanding per Capita (CR_OUT_PC), and Insurance Density (INS_DEN).

Separate PCAs were then estimated for each dimension to derive the latent scores used in the second stage of the hierarchical framework.

4.3.2 Stage 1 PCA: Construction of Dimension Scores

Separate Principal Component Analyses were performed for each dimension using the standardised indicators. For the penetration, availability and usage dimensions, the first principal component (PC1) was retained as the latent dimension score.

As reported in Table 4.2, PC1 recorded an eigenvalue greater than one for all three dimensions, satisfying the Kaiser criterion. It explained 96.31% of the variance in the penetration dimension, 91.70% in the availability dimension and 99.47% in the usage dimension, indicating that a single component adequately represents each conceptual dimension.

The corresponding component loadings are presented in Table 4.3. The penetration and usage dimensions exhibit positive loadings of similar magnitude across all indicators, indicating that each variable contributes positively to the underlying dimension. Within the availability dimension,

Scheduled Commercial Bank Offices, Rural Scheduled Commercial Bank Offices, Insurance Offices and Internet Subscribers recorded positive loadings, whereas the employee variable (EMP) recorded a negative loading.

This reflects the construction of EMP as bank employees per lakh deposit accounts. Because deposit accounts increased much faster than bank staffing during the study period, the employee-to-deposit ratio declined despite continued expansion in financial services. Consequently, EMP exhibits an inverse relationship with the remaining availability indicators, producing a negative loading in the first principal component. The variable was nevertheless retained because the negative loading reflects its empirical relationship with the latent availability dimension rather than weaker financial inclusion.

The retained first principal components constitute the penetration, availability and usage scores used in the second stage of the hierarchical PCA.

Table 4.2 Stage 1 PCA Results

Dimension	Eigenvalue	Variance Explained (%)	Component Retained
Penetration	1.93	96.31	PC1
Availability	4.58	91.70	PC1
Usage	2.98	99.47	PC1

Table 4.3 Stage 1 Component Loadings

Indicator	Penetration	Availability	Usage
DEP_ACC	0.707	—	—
CR_ACC	0.707	—	—
SCB_OFF	—	0.465	—
SCB_OFF_R	—	0.456	—
EMP	—	−0.451	—
INS_OFF	—	0.433	—
INT_SUB	—	0.430	—
DEP_PC	—	—	0.578
CR_OUT_PC	—	—	0.577
INS_DEN	—	—	0.576

4.3.3 Stage 2 PCA: Construction of the Composite Component

The three latent dimension scores obtained in Stage 1 were standardised and used as the inputs for a second Principal Component Analysis to derive a single component representing overall financial inclusion (Cámara & Tuesta, 2014).

As reported in Table 4.4, the first principal component (PC1) recorded an eigenvalue of 2.87, exceeding the Kaiser criterion of one, whereas the remaining components had eigenvalues below this threshold. PC1 explained 95.53 percent of the total variation in the three dimension scores, indicating that it adequately captures their common variation. Accordingly, only PC1 was retained.

The component loadings presented in Table 4.5 show that penetration, availability and usage all recorded positive loadings of similar magnitude, indicating that each dimension contributes positively and almost equally to overall financial inclusion. These loadings were subsequently used as the dimension weights for constructing the composite Financial Inclusion Index.

Table 4.4 Stage 2 PCA Results

Component	Eigenvalue	Variance Explained (%)	Cumulative (%)
PC1	2.866	95.53	95.53
PC2	0.130	4.32	99.85
PC3	0.005	0.15	100.00

Table 4.5 Stage 2 Component Loadings

Dimension	Loading
Penetration	0.5848
Availability	0.5641
Usage	0.5829

4.4 Construction and Validation of the Financial Inclusion Index

Following the two-stage PCA, the retained first principal component from Stage 2 was used to construct the Financial Inclusion Index (FII). Let P , A and U denote the latent scores for the penetration, availability and usage dimensions obtained from Stage 1, and let λ_p , λ_a and λ_u represent the corresponding Stage 2 loadings. The raw financial inclusion score was computed as:

$$Raw\ Index = \lambda_p P + \lambda_A A + \lambda_U U$$

Substituting the estimated loadings,

$$Raw\ Index = 0.5848P + 0.5641A + 0.5829U$$

Because principal component scores are centred around zero and may take negative values, the raw scores were transformed as:

$$Shifted\ Index = Raw\ Index + | \min(Raw\ Index) | + 1$$

This transformation ensured that all index values were positive while preserving the relative differences among observations. The shifted index was used as the final Financial Inclusion Index for the state-level analysis.

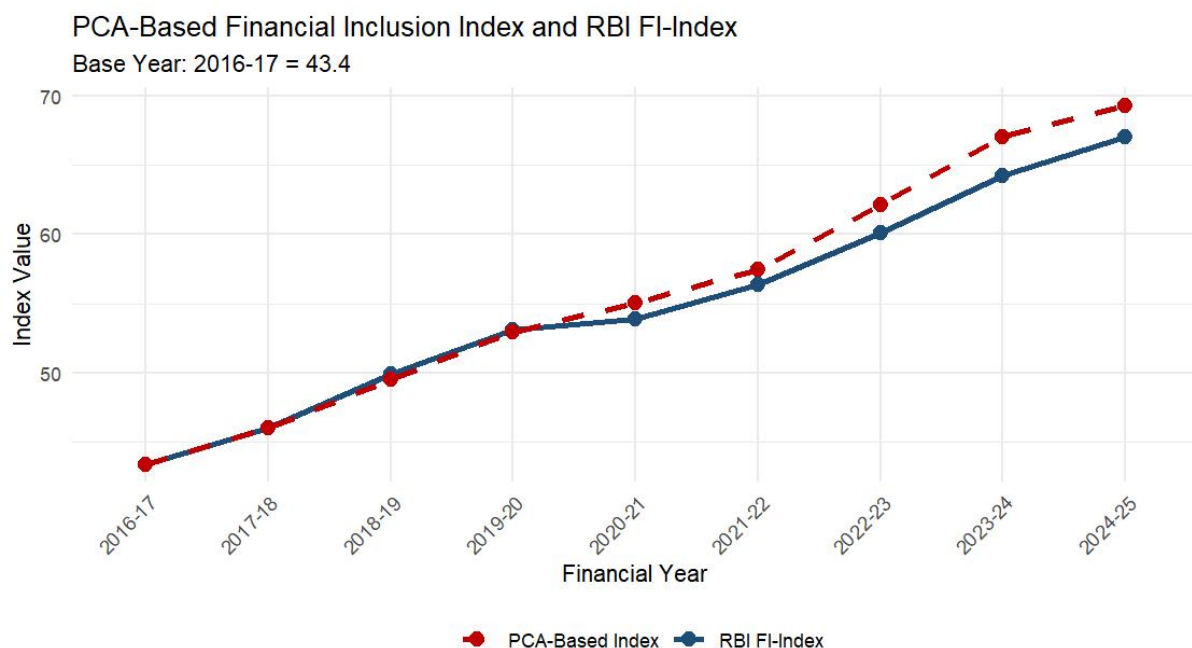
To validate the methodology, the index was first constructed for the national dataset covering 2004–05 to 2024–25. As the RBI Financial Inclusion Index is available only from 2016–17 onwards, the comparison was limited to 2016–17 to 2024–25. For this purpose, the national index was rebased by setting 2016–17 = 43.4, matching the RBI FI-Index. This rebasing was performed solely for validation and was not applied to the state-level index.

Table 4.6 and Figure 4.4 compare the PCA-based FII with the RBI FI-Index. The two series exhibit remarkably similar movements over the study period. The Pearson correlation coefficient is 0.9982 and the Spearman rank correlation coefficient is 1.000, indicating an almost perfect linear relationship and identical year-wise rankings. These findings provide strong empirical support for the validity of the proposed methodology

Table 4.6 Comparison of the Hierarchical PCA-Based Financial Inclusion Index and RBI FI-Index

Financial Year	PCA-Based FII	RBI FI-Index
2016–17	43.40	43.40
2017–18	46.00	46.00
2018–19	49.53	49.90
2019–20	52.93	53.10
2020–21	55.02	53.90
2021–22	57.49	56.40
2022–23	62.12	60.10
2023–24	67.00	64.20
2024–25	69.27	67.00

Figure 4.4 Comparison of the Two-Stage PCA-Based Financial Inclusion Index and RBI FI-Index



4.5 Robustness and Sensitivity Analysis

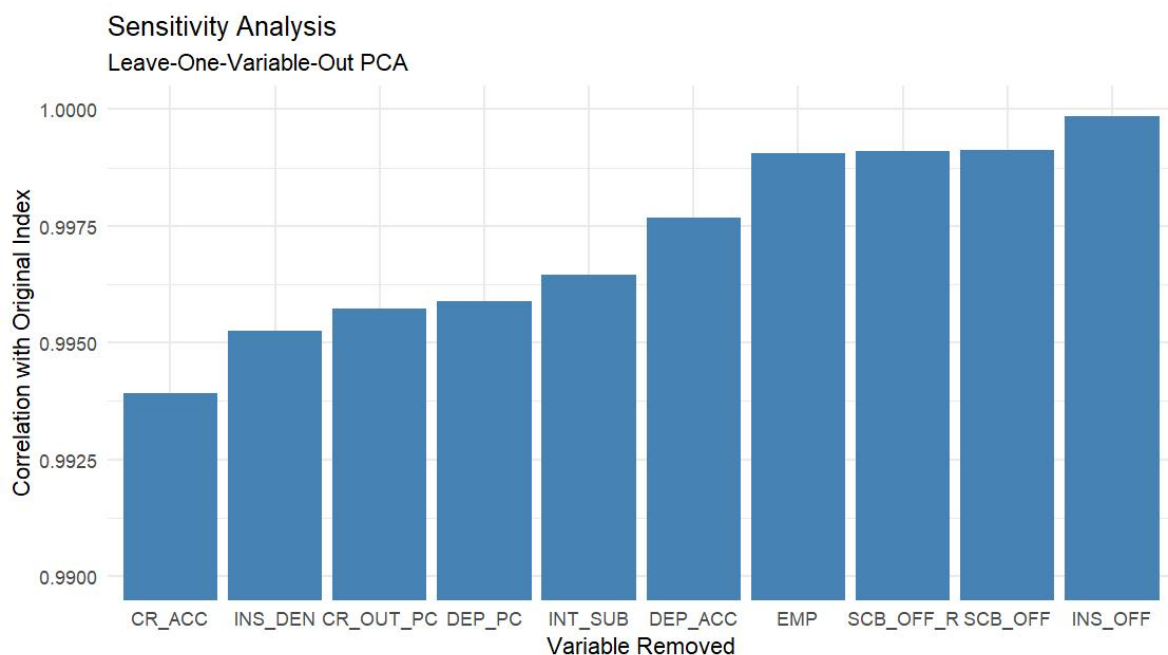
To assess the robustness of the proposed Financial Inclusion Index (FII), a leave-one-variable-out sensitivity analysis was conducted. The complete hierarchical two-stage PCA procedure was repeated ten times, each time excluding one indicator. The resulting indices were compared with the original index using Pearson and Spearman correlation coefficients (OECD, 2008).

The results, presented in Figure 4.5, confirm a high degree of robustness. Pearson correlation coefficients range from 0.9939 to 0.9999, while the Spearman rank correlation remains 1.000 across all specifications, indicating that excluding any single indicator produces only marginal changes and does not affect the relative ranking of observations.

Among the indicators, excluding Credit Accounts per lakh Population (CR_ACC) produced the lowest Pearson correlation (0.9939), whereas excluding Insurance Offices per lakh Population (INS_OFF) produced the highest (0.9999). Excluding Employees per lakh Deposit Accounts (EMP) also yielded a Pearson correlation above 0.999, confirming that its negative loading in the availability dimension does not distort the overall index.

Overall, the sensitivity analysis demonstrates that the hierarchical two-stage PCA framework is robust and that the Financial Inclusion Index is not unduly influenced by any individual indicator.

Figure 4.5 Leave-One-Variable-Out Sensitivity Analysis



4.6 Extension of the Two-Stage PCA Framework to the State Panel and State-wise Financial Inclusion Index

Following validation at the national level, the two-stage PCA framework was extended to construct Financial Inclusion Indices (FIIs) for a balanced panel of 28 Indian states covering 2016–17 to 2024–25. The state-level indicators were standardised using the pooled mean and standard deviation of the complete state panel, placing all state-year observations on a common statistical scale. This ensured that differences in the index reflected genuine variation in financial inclusion and enabled meaningful comparisons across states and over time.

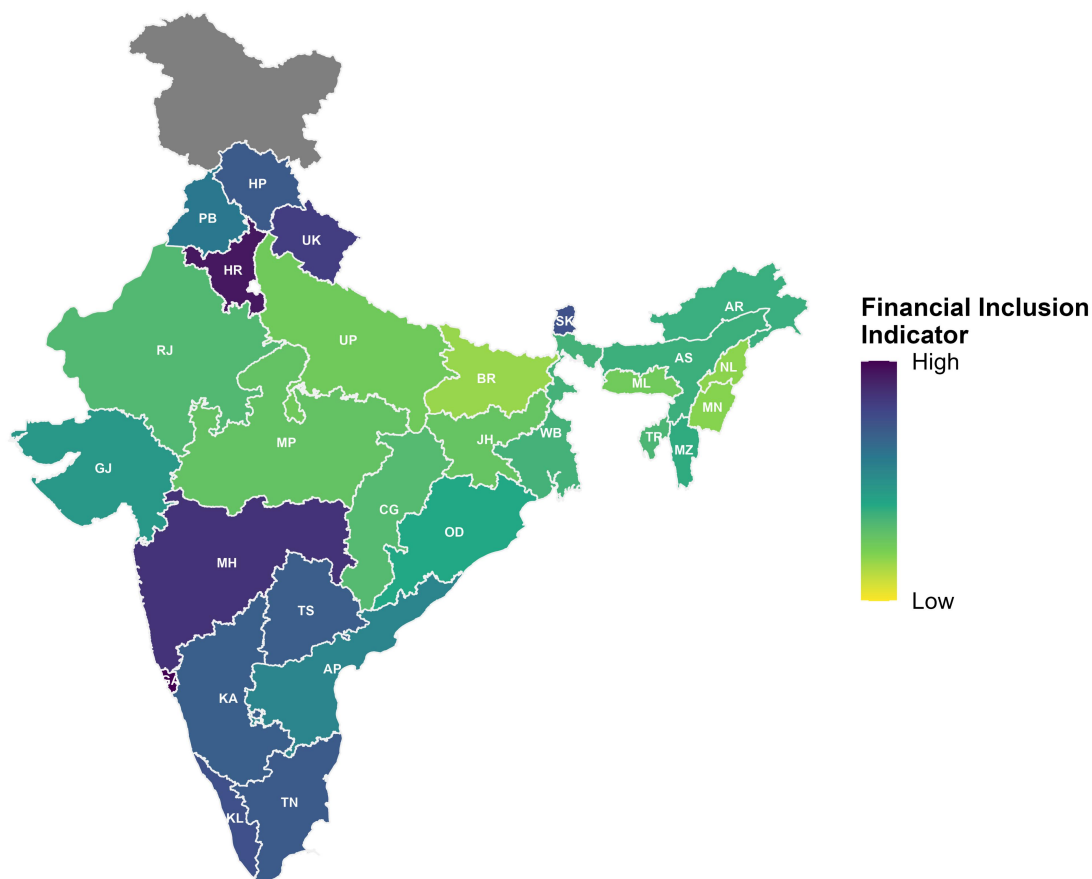
The standardised indicators were then combined using the nationally estimated two-stage PCA framework. In Stage 1, the national loadings were applied separately to the penetration, availability and usage indicators to derive dimension scores for each state-year observation. These scores were subsequently aggregated using the Stage 2 national loadings to obtain the final Financial Inclusion Index. Applying a common set of nationally derived loadings ensured that the index measured the same underlying construct consistently across all states and years.

Using this methodology, Financial Inclusion Indices were computed for all 28 states over 2016–17 to 2024–25. Figure 4.6 illustrates their distribution across Indian states in 2024–25. These indices form the basis for the interstate analysis presented in the following chapter.

Figure 4.6 State-wise Financial Inclusion Index across Indian States (2024–25)

Financial Inclusion across Indian States (2024-25)

Two-Stage PCA-Based Financial Inclusion Indicator



Source: Author's calculations

The PCA-based state-level indices developed in this chapter provide the first measure of interstate financial inclusion constructed under the hierarchical two-stage framework. To assess the robustness of these estimates, the following chapter constructs an alternative set of indices using Sarma's distance-based methodology.

5. SARMA METHODOLOGY

5.1 Overview of the Sarma Methodology

As discussed in Chapter 2, Sarma (2008, 2012) proposed one of the earliest multidimensional approaches for measuring financial inclusion using the dimensions of penetration, availability and usage. Alongside the hierarchical PCA framework, this study applies the Sarma methodology to construct an alternative Financial Inclusion Index (FII) for the selected Indian states.

Unlike Principal Component Analysis (PCA), which derives weights statistically from the data, the Sarma methodology uses a distance-based approach to combine predefined dimensions into a composite index. Its conceptual simplicity, transparency and widespread use in the financial inclusion literature make it an appropriate benchmark for comparison.

Following Sarma (2008, 2012), the selected indicators were grouped into the three dimensions, normalised to a common scale, used to construct dimension indices and subsequently aggregated into the overall Financial Inclusion Index.

5.2 Normalisation of Financial Inclusion Indicators

Because the selected indicators are measured in different units, they were normalised to a common scale ranging from 0 to 1, consistent with Sarma (2008, 2012). Here, 0 represents the lowest observed level of financial inclusion and 1 the upper benchmark, ensuring comparability while preserving relative variation.

The normalised value of indicator i is computed as:

$$d_i = \frac{A_i - m_i}{M_i - m_i}$$

where A_i denotes the observed value, m_i the lower bound and M_i the upper bound..

Following Sarma (2012), the lower bound was taken as the minimum observed value and the upper bound as the 94th percentile of the pooled state panel rather than the observed maximum. This reduces the influence of extreme observations while providing a more realistic benchmark for financial inclusion. Values above the upper bound were assigned a value of 1, whereas those below the lower bound were assigned 0.

The normalised indicators were then grouped into the penetration, availability and usage dimensions to construct the corresponding dimension indices.

5.3 Construction of the Dimension Indices

Following normalisation, the indicators were grouped into the three dimensions of penetration, availability and usage proposed by Sarma (2008, 2012). Consistent with the original methodology, equal weights were assigned to all indicators within each dimension, and each dimension index was calculated as the arithmetic mean of the corresponding normalised indicators.

The three dimensions comprise the same ten indicators used in the hierarchical PCA framework, ensuring direct comparability between the Sarma- and PCA-based indices. The resulting dimension indices range from 0 to 1, with higher values indicating better financial inclusion, and were subsequently combined using Sarma's distance-based methodology to construct the overall Financial Inclusion Index.

Following the construction of the three dimension indices, the overall Financial Inclusion Index (FII) was computed using the distance-based methodology proposed by Sarma (2008, 2012). The index considers both the distance from the worst possible situation, where all dimension indices equal zero, and the distance from the ideal situation, where all dimension indices equal one. This ensures that the index reflects both the overall level of financial inclusion and the balance across the three dimensions.

The distance from the worst point is computed as:

$$X_1 = \frac{\sqrt{P^2 + A^2 + U^2}}{\sqrt{3}}$$

where P, A and U denote the penetration, availability and usage dimension indices, each ranging from 0 to 1 under the Sarma framework and distinct from the unbounded PCA dimension scores presented in Chapter 4. The distance from the ideal point is computed as:

$$X_2 = 1 - \frac{\sqrt{(1-P)^2 + (1-A)^2 + (1-U)^2}}{\sqrt{3}}$$

The final Financial Inclusion Index is calculated as:

$$FII = \frac{X_1 + X_2}{2}$$

The resulting index ranges from 0 to 1, with higher values indicating greater financial inclusion. The Sarma-based Financial Inclusion Index was computed for each state-year observation.

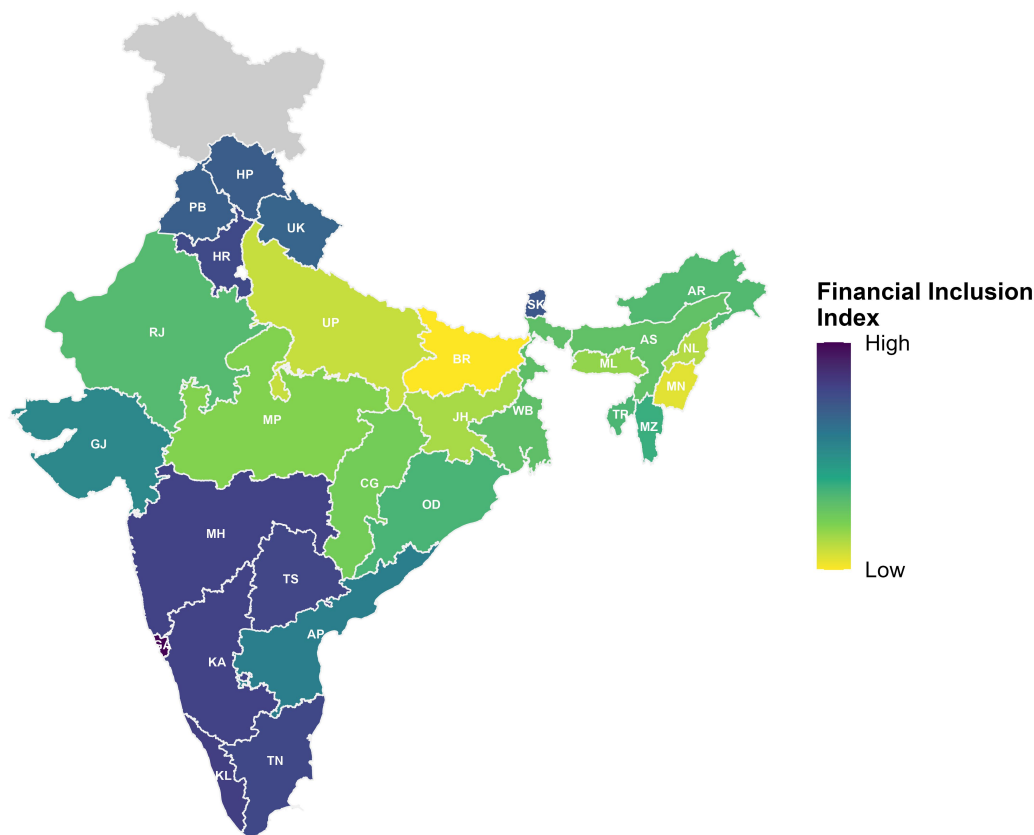
5.4 State-wise Sarma Financial Inclusion Index

Using the methodology described in this chapter, the Sarma-based Financial Inclusion Index was computed for a balanced panel of 28 Indian states covering 2016–17 to 2024–25. Figure 5.1 presents the distribution of financial inclusion across states in 2024–25.

Figure 5.1 State-wise Sarma Financial Inclusion Index (2024–25)

Financial Inclusion across Indian States (2024-25)

Sarma (2012) Distance-Based Financial Inclusion Index



Source: Author's calculations

6. RESULTS AND ANALYSIS OF FINANCIAL INCLUSION ACROSS INDIAN STATES

This chapter presents the empirical findings of the study. It examines state-level financial inclusion and its spatial distribution, temporal changes in interstate rankings, regional patterns, comparison with the Sarma methodology, the contribution of the penetration, availability and usage dimensions through a Financial Inclusion Development Typology, and evidence on convergence and interstate mobility.

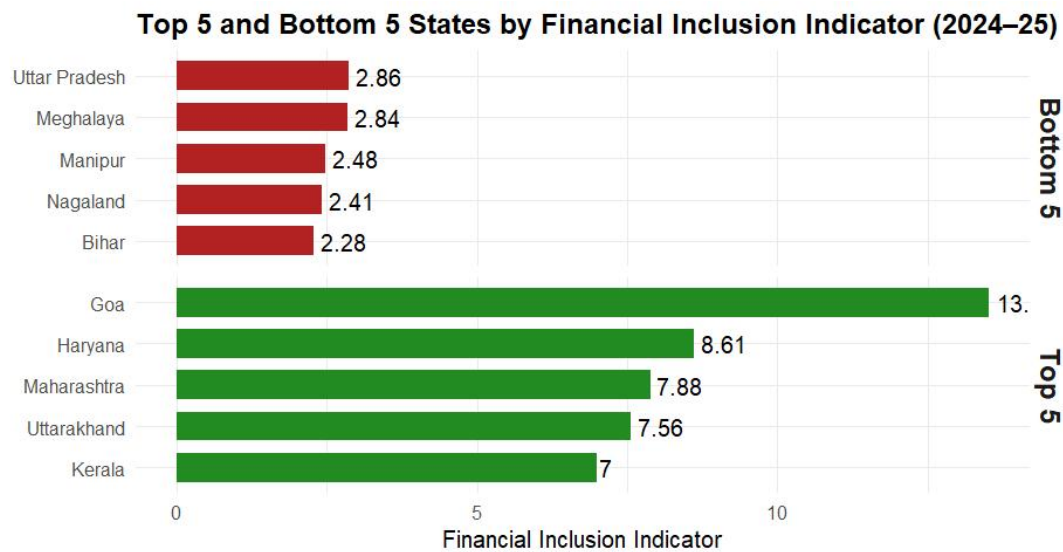
6.1 State-wise Levels and Spatial Distribution of Financial Inclusion

6.1.1 State-wise Financial Inclusion Rankings

The state rankings reveal substantial and persistent disparities in financial inclusion across India. In 2024–25, Goa ranked first, followed by Haryana and Maharashtra, while Bihar ranked last, preceded by Nagaland and Manipur. Goa remained the highest-ranked state throughout the study period.

Although the overall hierarchy remained broadly stable, a few states experienced notable upward or downward movement, discussed in Section 6.2.2, whereas states such as Jharkhand and Bihar showed limited mobility and remained near the bottom. In general, the rankings indicate a persistent interstate hierarchy with selective changes over time. The associated spatial pattern is examined in the next section.

Figure 6.1: Top Five and Bottom Five States by Financial Inclusion Index, 2024–25



Source: Author's calculations.

6.1.2 Spatial Distribution of Financial Inclusion

Figure 4.6 shows that financial inclusion follows a distinct geographical pattern rather than a random distribution. Higher financial inclusion is concentrated primarily in the western and southern regions, led by Goa, Maharashtra, Kerala and Tamil Nadu, whereas lower financial inclusion is concentrated across much of eastern and central India, particularly in Bihar, Jharkhand, Uttar Pradesh, Madhya Pradesh and Chhattisgarh. Within the North-Eastern region, most states record relatively similar levels of financial inclusion despite lower average index values, although Arunachal Pradesh, Assam and Mizoram perform somewhat better than Meghalaya, Manipur, Nagaland and Tripura.

Overall, the spatial distribution indicates broad regional clustering alongside notable interstate differences, suggesting that regional averages conceal substantial within-region variation. These patterns provide the basis for the regional analysis in Section 6.3.

6.2 Temporal Dynamics of Financial Inclusion

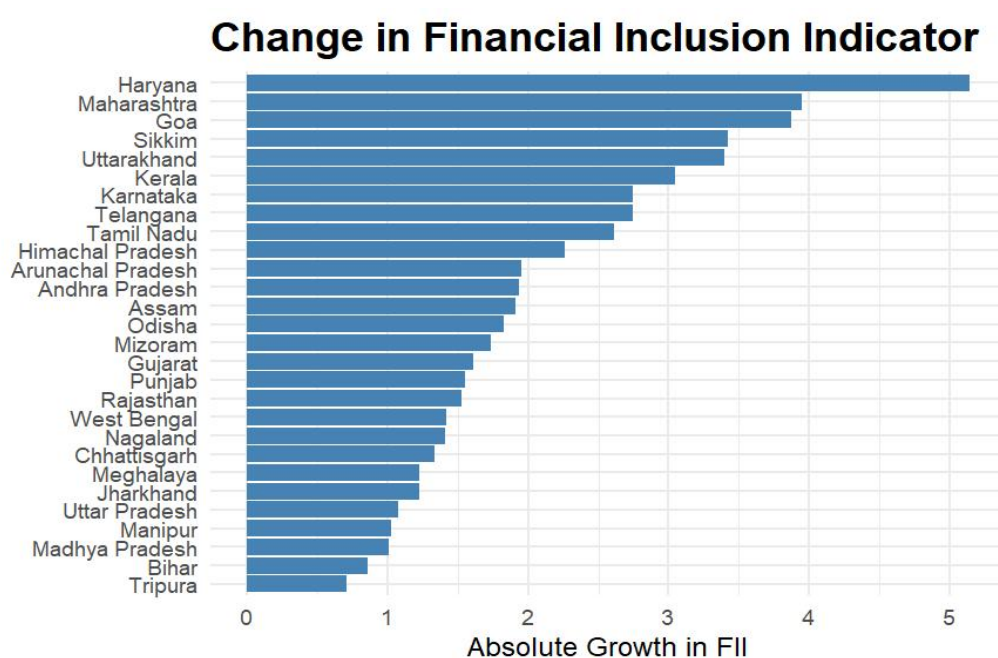
6.2.1 Growth in State-level Financial Inclusion

Figure 6.2 presents the absolute change in the Financial Inclusion Index (FII) across Indian states between 2016–17 and 2024–25. Financial inclusion improved in every state during the study period, although the magnitude of improvement varied substantially, ranging from **0.71** index points in Tripura to 5.14 in Haryana.

The largest gains were recorded by Haryana (5.14), Maharashtra (3.95), Goa (3.87), Sikkim (3.42) and Uttarakhand (3.40). Kerala, Karnataka, Telangana and Tamil Nadu also recorded increases exceeding 2.5 points. In contrast, Tripura, Bihar, Madhya Pradesh, Manipur and Uttar Pradesh recorded increases of less than 1.1 points, while Jharkhand, Meghalaya and Chhattisgarh also experienced comparatively modest gains. These findings indicate that financial inclusion expanded nationwide but at markedly different rates across states.

Given the construction of the index, the analysis focuses on absolute rather than percentage changes. Because the PCA scores were translated by a constant to obtain positive index values, percentage growth depends partly on an arbitrary origin and is therefore not directly comparable across states. Absolute changes preserve the underlying differences in PCA scores and provide a more appropriate basis for comparison. Whether these differences altered the interstate hierarchy is examined in the following section.

Figure 6.2: Absolute change in the Financial Inclusion Index (FII) across Indian states between 2016–17 and 2024–25



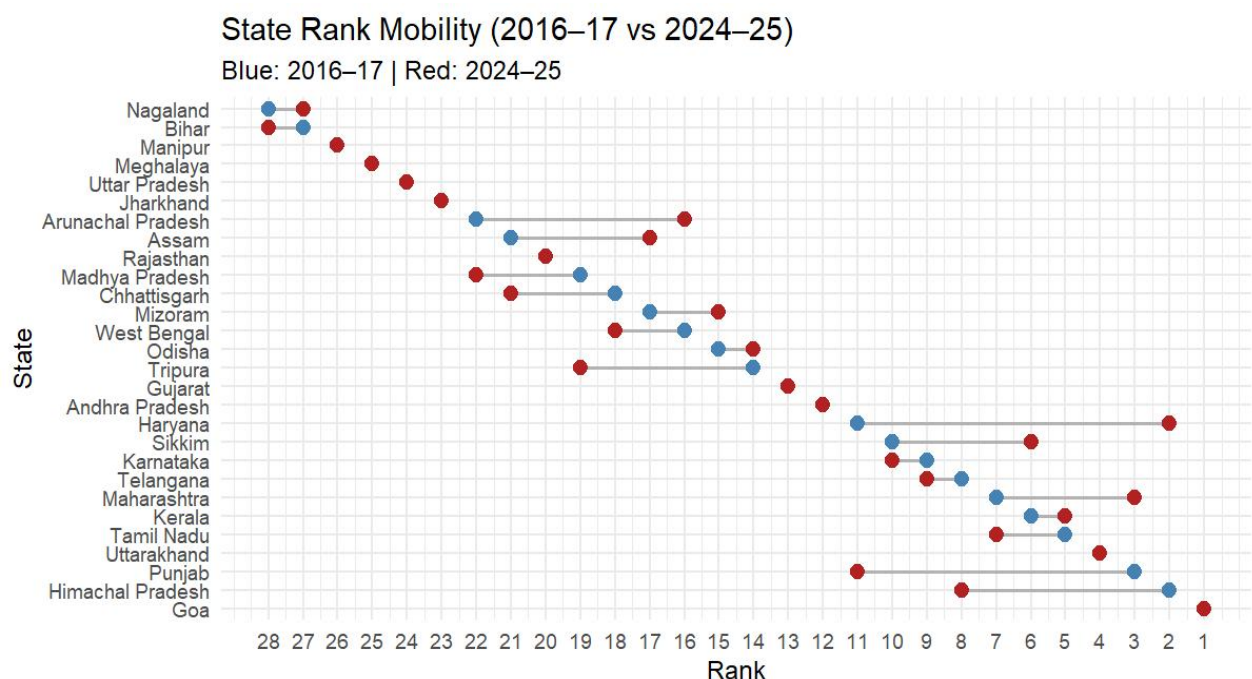
6.2.2 Evolution of State Rankings and Rank Mobility

Although all states recorded higher Financial Inclusion Index values over the study period, these improvements produced only limited changes in their relative rankings. The Spearman rank correlation coefficient between 2016–17 and 2024–25 was 0.911 ($p < 0.001$), indicating a strong and statistically significant association between initial and final rankings. This finding is supported by the Wilcoxon signed-rank test, which detected no significant change in the overall rank distribution ($p = 0.952$). These results indicate that while individual states experienced some rank mobility, the interstate hierarchy remained largely unchanged.

Despite this overall stability, a few states recorded notable improvements in their relative positions. Haryana showed the greatest upward mobility, rising from 11th to 2nd place. Arunachal Pradesh moved from 22nd to 16th, while Maharashtra, Sikkim and Assam each gained four positions. Smaller improvements were recorded by Mizoram, Kerala, Odisha and Nagaland, reflecting faster improvements in financial inclusion relative to other states. Conversely, Punjab recorded the largest decline, falling from 3rd to 11th place, followed by Himachal Pradesh (2nd to 8th) and Tripura (14th to 19th). Chhattisgarh and Madhya Pradesh each declined by three positions, while Tamil Nadu and West Bengal fell by two. These declines reflect slower improvements relative to other states rather than reductions in financial inclusion.

Overall, the rank mobility analysis indicates substantial persistence in interstate rankings. Nine states retained the same rank throughout the study period, while most others experienced only modest changes.

Figure 6.3: State Rank Mobility



6.3 Regional Patterns in Financial Inclusion

6.3.1 Regional Trends and Disparities

Having examined state-level changes, the analysis now considers whether financial inclusion evolved differently across India's six geographical regions. Figures 6.4 and 6.5 present the average state-level Financial Inclusion Index between 2016–17 and 2024–25 and the regional rankings in the final year. Financial inclusion improved across all regions, although the regional hierarchy changed little over time.

The Western region consistently recorded the highest average Financial Inclusion Index and the largest absolute improvement. The Southern region remained second throughout the study period, while the Northern region grew rapidly and substantially narrowed the gap with the South by 2024–25. Together, these regions formed the upper tier of financial inclusion.

The Central, Eastern and North-Eastern regions remained below the leading regions despite steady improvements. By 2024–25, the Central and Eastern regions occupied intermediate positions, whereas the North-Eastern region continued to record the lowest average financial inclusion. Overall, financial inclusion expanded across all regions, but persistent regional differences remained a defining feature of India's financial landscape. The extent of variation within regions is examined in the following subsection.

Figure 6.4 Regional Trends in the Financial Inclusion Index (2016–17 to 2024–25)

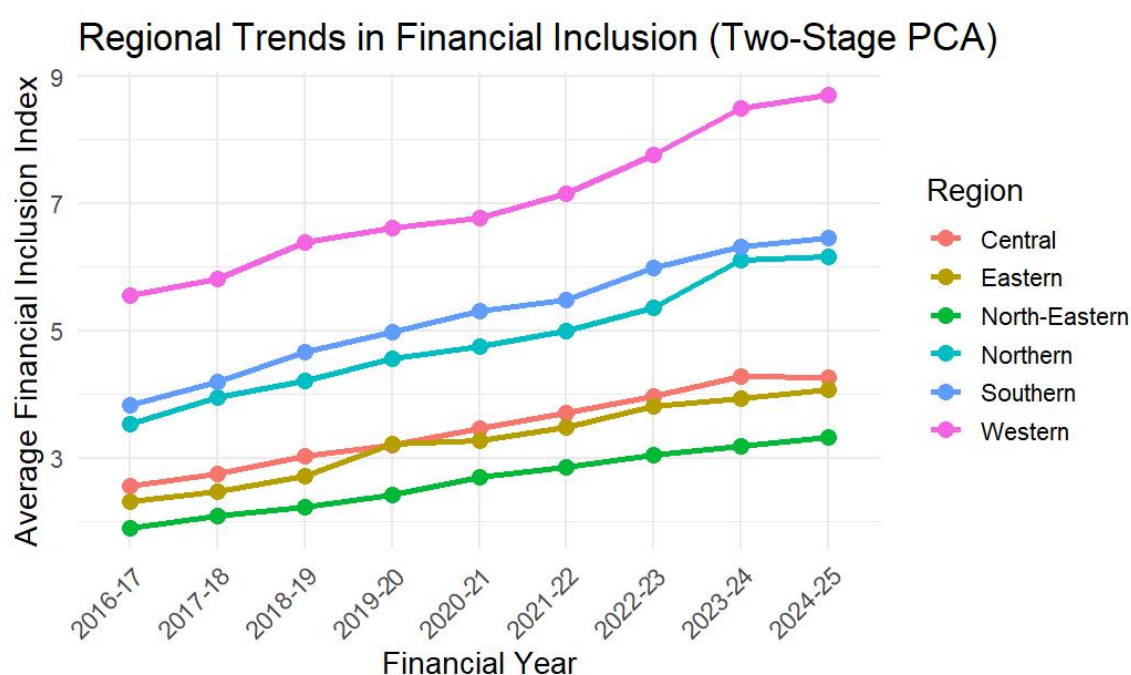
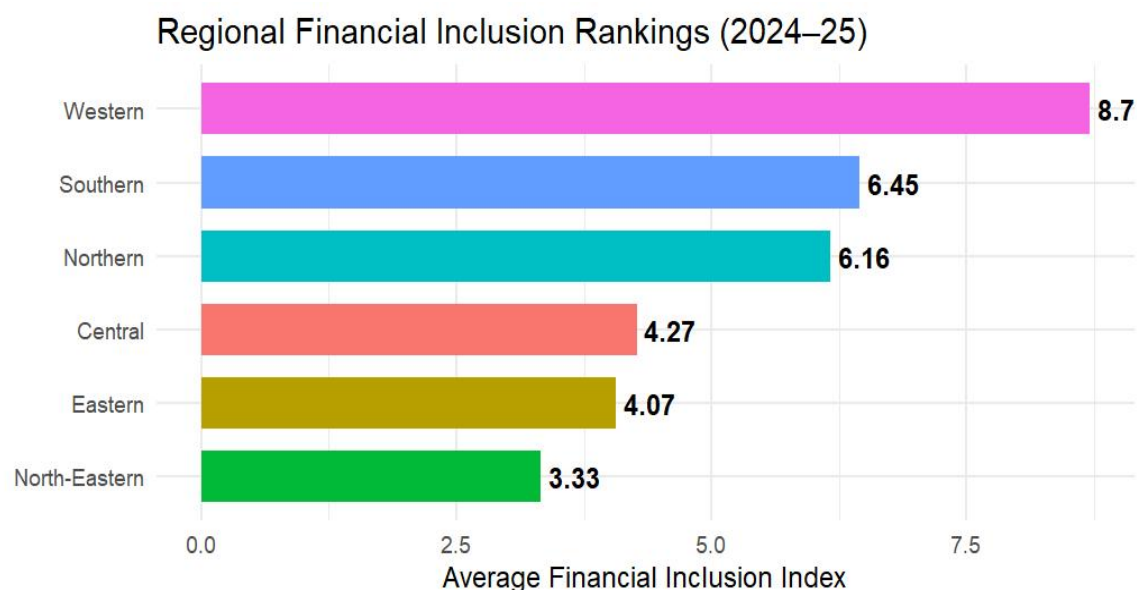


Figure 6.5 Regional Rankings of the Financial Inclusion Index (2024–25)



6.3.2 Within-Region Heterogeneity

Regional averages conceal substantial variation among states within the same region. Table 6.1 reports both the standard deviation (SD) and the coefficient of variation (CV) of the Financial Inclusion Index in 2024–25. SD measures absolute dispersion, whereas CV measures relative dispersion, enabling comparisons across regions with different average levels of financial inclusion.

The Western region recorded the highest standard deviation (4.46), reflecting substantial absolute disparities, largely driven by Goa's exceptionally high financial inclusion. However, the Central region exhibited the highest relative inequality, with the largest coefficient of variation (0.519), marginally exceeding the Western region (0.512). The Northern and Eastern regions showed moderate dispersion, while the Southern region recorded both the lowest SD (0.63) and CV (0.098), indicating consistently high and balanced financial inclusion. Although the North-Eastern region had the lowest average financial inclusion, its relatively low SD (0.73) and CV (0.218) suggest comparatively homogeneous performance across states.

Southern region remained the most homogeneous throughout the study period, while the Western region became less dispersed over time but continued to exhibit substantial inequality. The Eastern region became progressively more unequal, the Northern region maintained moderate heterogeneity, and the North-Eastern region became increasingly homogeneous, with its CV declining from 0.331 in 2016–17 to 0.218 in 2024–25.

To assess whether these regional differences were statistically significant, the 2024–25 Financial Inclusion Index values were analysed using the Kruskal–Wallis test, which identified significant regional differences ($\chi^2 = 12.52$, $p = 0.028$). A one-way ANOVA produced consistent results ($F = 4.28$, $p = 0.007$), confirming statistically significant regional disparities in financial inclusion.

Table 6.1 Coefficient of Variation of the Financial Inclusion Index by Region (2024–25)

Region	Mean FII (2024–25)	Coefficient of Variation	Standard Deviation	Interpretation
Western	8.71	0.512	4.4607	Highest-performing but internally unequal
Southern	6.45	0.098	0.6302	High-performing and balanced
Northern	6.16	0.340	2.0966	Moderate internal inequality
Central	4.27	0.519	2.2129	Highest internal inequality
Eastern	4.07	0.438	1.7818	Moderate heterogeneity
North-Eastern	3.33	0.218	0.7256	Low-performing but relatively homogeneous

6.3.3 Spatial Autocorrelation Analysis

The regional analysis established substantial geographical disparities in financial inclusion across India. However, regional averages do not indicate whether neighbouring states exhibit similar levels of financial inclusion or whether the observed patterns arise from spatial dependence. To examine the spatial structure of financial inclusion, spatial autocorrelation was assessed using Global Moran's I, which measures overall spatial clustering, and Local Moran's I (LISA), which identifies statistically significant local clusters and spatial outliers. A Queen contiguity spatial weights matrix was used to define neighbouring states.

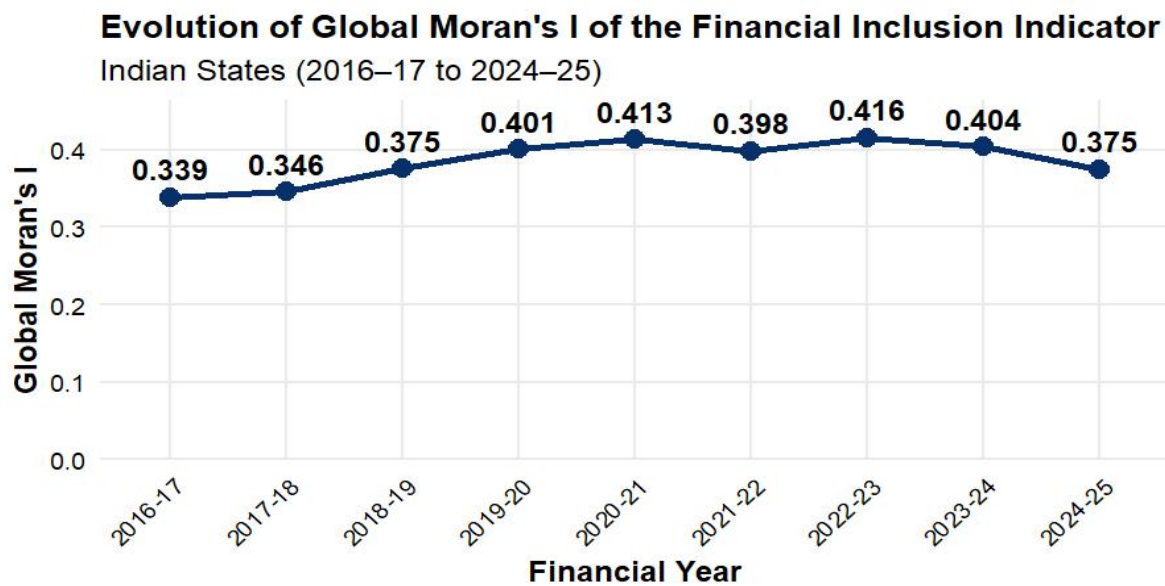
6.3.3.1 Global Moran's I

Table 6.2. Global Moran's I Statistics for the Financial Inclusion Index (2016–17 to 2024–25)

Financial Year	Moran's I	Expected Moran's I	Z-statistic	p-value	Significance
2016–17	0.339	-0.037	3.38	<0.001	***
2017–18	0.346	-0.037	3.34	<0.001	***
2018–19	0.375	-0.037	3.61	<0.001	***
2019–20	0.401	-0.037	3.61	<0.001	***
2020–21	0.413	-0.037	3.80	<0.001	***
2021–22	0.398	-0.037	3.65	<0.001	***

Financial Year	Moran's I	Expected Moran's I	Z-statistic	p-value	Significance
2022–23	0.416	-0.037	3.69	<0.001	***
2023–24	0.404	-0.037	3.54	<0.001	***
2024–25	0.375	-0.037	3.27	<0.001	***

Figure 6.6. Evolution of Global Moran's I for the Financial Inclusion Index across Indian States (2016–17 to 2024–25)



Note: Under the null hypothesis of spatial randomness, the expected value of Global Moran's I is -0.037 . The consistently higher observed values indicate persistent positive spatial autocorrelation throughout the study period.

Table 6.2 and Figure 6.6 show that Global Moran's I remained positive and statistically significant throughout the study period, increasing from 0.339 in 2016–17 to a peak of 0.416 in 2022–23 before moderating to 0.375 in 2024–25. In every year, the observed Moran's I exceeded its expected value under spatial randomness (-0.037), with z-statistics above 3 and $p < 0.001$, leading to rejection of the null hypothesis of spatial randomness. These findings indicate that neighbouring states consistently exhibited similar levels of financial inclusion rather than a random spatial distribution.

Spatial clustering strengthened gradually for most of the study period before moderating slightly in the final two years. Nevertheless, the consistently positive Moran's I values demonstrate that geographical proximity remained an important determinant of interstate financial inclusion throughout 2016–17 to 2024–25. Together with the regional analysis,

these findings indicate that interstate disparities reflect both broad regional differences and significant spatial dependence.

6.3.3.2 Local Moran's I (LISA)

Table 6.3. Statistically Significant LISA Clusters (2024–25)

LISA Cluster Type	States	Interpretation
High–High (HH)	Karnataka	State with a high Financial Inclusion Index surrounded by neighbouring states with similarly high financial inclusion.
Low–Low (LL)	Jharkhand, Assam	States with relatively low financial inclusion surrounded by neighbouring states with similarly low financial inclusion.
High–Low (HL)	None	No statistically significant high-performing state surrounded by low-performing neighbours.
Low–High (LH)	None	No statistically significant low-performing state surrounded by high-performing neighbours.

While Global Moran's I measures overall spatial dependence, Local Moran's I identifies where statistically significant clustering occurs. Only three significant local clusters were detected. Karnataka emerged as the sole High–High cluster, indicating high financial inclusion surrounded by neighbouring states with similarly high levels. In contrast, Jharkhand and Assam formed the only significant Low–Low clusters, representing concentrations of relatively low financial inclusion. These findings are consistent with the earlier regional analysis, which identified stronger financial inclusion in parts of southern India and lower levels across sections of eastern and north-eastern India.

No statistically significant High–Low or Low–High spatial outliers were identified. This indicates that highly financially included states are generally not surrounded by substantially lower-performing neighbours, and vice versa. Instead, spatial dependence is characterised primarily by neighbouring states with similar levels of financial inclusion. The limited number of significant local clusters suggests that the positive Global Moran's I reflects a broad spatial pattern rather than numerous isolated hotspots.

Overall, Global Moran's I confirms persistent positive spatial dependence throughout the study

period, while Local Moran's I shows that statistically significant clustering is confined to a small number of locations. Together, these findings suggest that neighbouring states often share similar financial inclusion outcomes, likely reflecting common socioeconomic conditions, institutional characteristics, geographical proximity and regional spillover effects, although statistically significant local clustering remains limited.

6.4 Comparison of the Two-Stage PCA and Sarma Indices

6.4.1 Purpose and Scope

To assess the robustness of the findings, the Financial Inclusion Index constructed using the Two-Stage Principal Component Analysis (PCA) is compared with the Sarma distance-based index. Rather than viewing the two methods as competing approaches, the comparison evaluates whether the main findings remain consistent under alternative methods of normalisation, weighting and aggregation. Specifically, it examines the agreement between the indices, the correspondence of state rankings, the identification of leading and lagging states, and the consistency of key findings, including the regional hierarchy of financial inclusion and patterns of interstate dispersion. The Two-Stage PCA remains the primary index throughout the study, while the Sarma index serves as an alternative specification for robustness assessment.

6.4.2 Overall Agreement and Temporal Stability

The Two-Stage PCA and Sarma indices exhibit strong overall agreement. Across the pooled panel of 252 state-year observations, the Pearson correlation is 0.919 ($p < 0.001$) and the Spearman correlation is 0.979 ($p < 0.001$). Because these pooled correlations reflect both cross-sectional differences among states and common changes over time, rank agreement was also evaluated using state rankings constructed separately for each year. The pooled within-year rankings likewise show strong correspondence, with a Spearman's rho of 0.947 and Kendall's tau of 0.828.

The agreement remained stable throughout the study period. Year-wise Pearson correlations ranged from 0.905–0.922, Spearman's rho from 0.932–0.956, and Kendall's tau from 0.794–0.847. The limited variation in these coefficients indicates that the correspondence between the two methodologies was consistently strong rather than driven by any single year. Despite their different weighting and aggregation procedures, the Two-Stage PCA and Sarma methodologies produced highly consistent index values and state rankings. The remaining differences in state rankings are examined in the following section.

Table 6.4 Overall and Year-wise Agreement between the Two-Stage PCA and Sarma Indices

Period	Pearson r	Spearman ρ	Kendall τ
Pooled index values (2016–17 to 2024–25)	0.919	0.979	—
Pooled within-year ranks (2016–17 to 2024–25)	—	0.947	0.828
2016–17	0.913	0.941	0.794
2017–18	0.916	0.950	0.815
2018–19	0.905	0.949	0.825
2019–20	0.922	0.932	0.799
2020–21	0.912	0.956	0.847
2021–22	0.909	0.946	0.841
2022–23	0.915	0.946	0.836
2023–24	0.912	0.953	0.836
2024–25	0.911	0.950	0.831

Note: Pooled index-value correlations are calculated across all 252 state-year observations. Pooled within-year rank correlations are based on state rankings constructed separately for each financial year. Annual Spearman coefficients are identical whether calculated from the index values or the corresponding within-year rankings.

6.4.3 Rank Correspondence and Extent of Divergence

Although Section 6.4.2 established strong agreement between the two indices, they do not produce identical rankings for every state. Across the 252 state-year observations, exact rank agreement occurred in 20.6% of cases, while 54.8% differed by no more than one position, 83.3% remained within three positions and 94.0% within five positions. The mean absolute rank difference was 1.87 positions, the median was one position and the maximum difference was nine positions.

Rank correspondence varied across states, indicating that disagreement was concentrated among a small number rather than widespread. Goa recorded complete agreement, ranking first under both methods in all nine years. Gujarat, Tamil Nadu, Andhra Pradesh, West Bengal and Punjab also showed average absolute rank differences below one position. In contrast, Uttarakhand, Himachal Pradesh, Meghalaya, Maharashtra, Haryana and Nagaland exhibited the largest average differences, indicating greater sensitivity to the choice of index construction method.

These differences reflect the distinct aggregation procedures of the two approaches, with the Two-Stage PCA deriving weights from the covariance structure of the data, whereas the Sarma method combines normalised dimensions using a bounded distance-based framework. Despite these methodological differences, the generally small rank deviations indicate that the overall interstate pattern of financial inclusion is robust to the choice of methodology.

Table 6.5: Summary of Rank Correspondence between the Hierarchical Two-Stage PCA and Sarma Indices

Measure	Result
Total state-year observations	252
Exact rank agreement (%)	20.63
Observations within 1 rank (%)	54.76
Observations within 3 ranks (%)	83.33
Observations within 5 ranks (%)	94.05
Mean absolute rank difference	1.87
Median absolute rank difference	1.00
Maximum absolute rank difference	9

Note: Rank difference is calculated as the rank assigned by the Two-Stage PCA index minus the rank assigned by the Sarma index. Absolute rank differences measure the extent of divergence irrespective of direction.

Despite these differences, the identification of leading and lagging states remained largely consistent in 2024–25. Three of the five highest-ranked states and four of the five lowest-ranked states were common to both methods, with Goa ranked first and Bihar last under both, indicating stronger agreement at the lower end of the distribution and suggesting that the identification of states with the greatest financial inclusion deficits is largely unaffected by the choice of index construction method.

Robustness of Regional Hierarchy

The regional hierarchy identified by the Two-Stage PCA index is closely reproduced by the Sarma index. Both methods ranked the Western, Southern and Northern regions first, second and third, respectively, while the North-Eastern region remained last. The only difference was the reversal in the ordering of the closely ranked Central and Eastern regions in 2024–25. Overall, the findings support a stable regional hierarchy of Western > Southern > Northern > {Central, Eastern} > North-Eastern, indicating that the broad pattern of regional disparities is robust to the choice of index construction method.

6.5 Dimensional Analysis

6.5.1 Performance across the Three Dimensions of Financial Inclusion

The hierarchical Two-Stage PCA generates separate scores for Penetration, Availability and Usage, allowing their individual contributions to be examined. Figures 6.7 and 6.8 present the national and regional performance across these three dimensions.

At the national level, the dimensions followed distinct trajectories. In 2016–17, Availability was the strongest dimension, substantially exceeding Penetration and Usage, reflecting the earlier expansion of financial infrastructure and access. By 2024–25, however, Usage recorded the largest increase and became the leading dimension, followed by Penetration, while Availability grew more modestly. This shift suggests that recent improvements in financial inclusion have been driven increasingly by greater use of formal financial services, whereas expansion of physical and institutional access has slowed.

Regional patterns broadly mirrored the national evidence but with notable differences. The Western region consistently recorded the highest scores across all three dimensions, with a particularly strong advantage in Usage after 2020–21. The Southern region also performed strongly, especially in Penetration, while the Northern region displayed a relatively balanced profile. In contrast, the Central, Eastern and North-Eastern regions remained below the national average across most dimensions throughout the study period, indicating broad-based weaknesses rather than deficiencies in any single dimension.

Overall, interstate differences in financial inclusion reflect variation in the relative strength of Penetration, Availability and Usage, as well as differences in overall index values. These findings provide the basis for the state-level dimensional analysis in the following subsection.

Figure 6.7 Performance Across the Three Dimensions of Financial Inclusion (India)

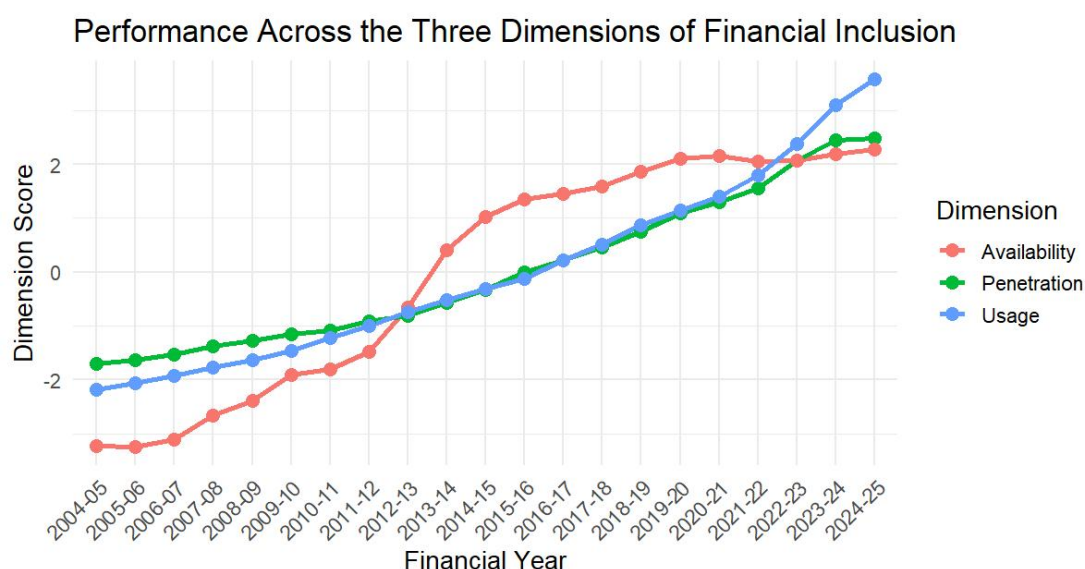
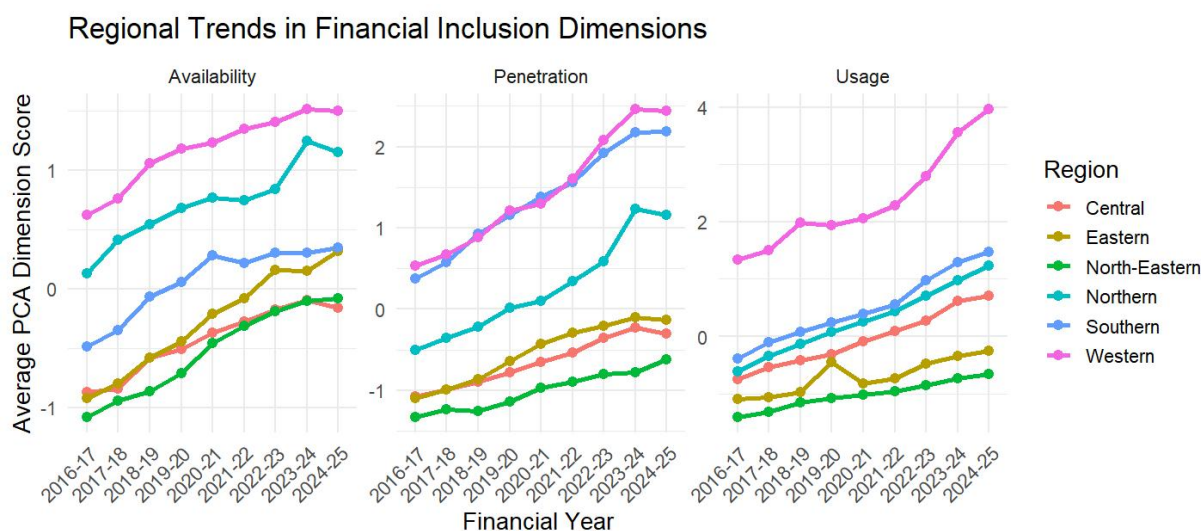


Figure 6.8 Regional Trends in Financial Inclusion Dimensions



6.5.2 Drivers of State-level Financial Inclusion Improvement

To identify the sources of improvement in financial inclusion, the change in each state's Financial Inclusion Index (FII) between 2016–17 and 2024–25 was decomposed into the weighted contributions of Penetration, Availability and Usage using the Stage 2 PCA loadings. Since the three dimension weights were broadly similar, the dominant driver primarily reflects differences in state performance rather than the weighting scheme. Figure 6.9 shows that similar improvements in the composite index often resulted from different combinations of the three dimensions.

The decomposition also explains the rank mobility discussed in Section 6.2.2. Haryana, which recorded the largest increase in both the Financial Inclusion Index and state ranking, achieved broad-based growth, with Penetration (40.4%) and Usage (37.2%) contributing almost equally. Maharashtra's improvement was driven mainly by Penetration, Sikkim's by Availability, and Arunachal Pradesh's by Usage. Assam also recorded Penetration-led growth, while Goa, despite retaining the highest rank throughout the study period, continued to derive the largest share of its improvement from Usage, indicating further deepening of financial service utilisation.

Three broad growth patterns emerge across the states. Usage was the dominant driver in Goa, Uttarakhand, Karnataka, Telangana, Gujarat, Rajasthan, Punjab and Himachal Pradesh. Uttarakhand was particularly notable, with nearly two-thirds of its improvement attributable to Usage, the highest share among all states. This indicates that recent gains in many states were driven increasingly by greater utilisation of existing financial services rather than continued expansion of access alone.

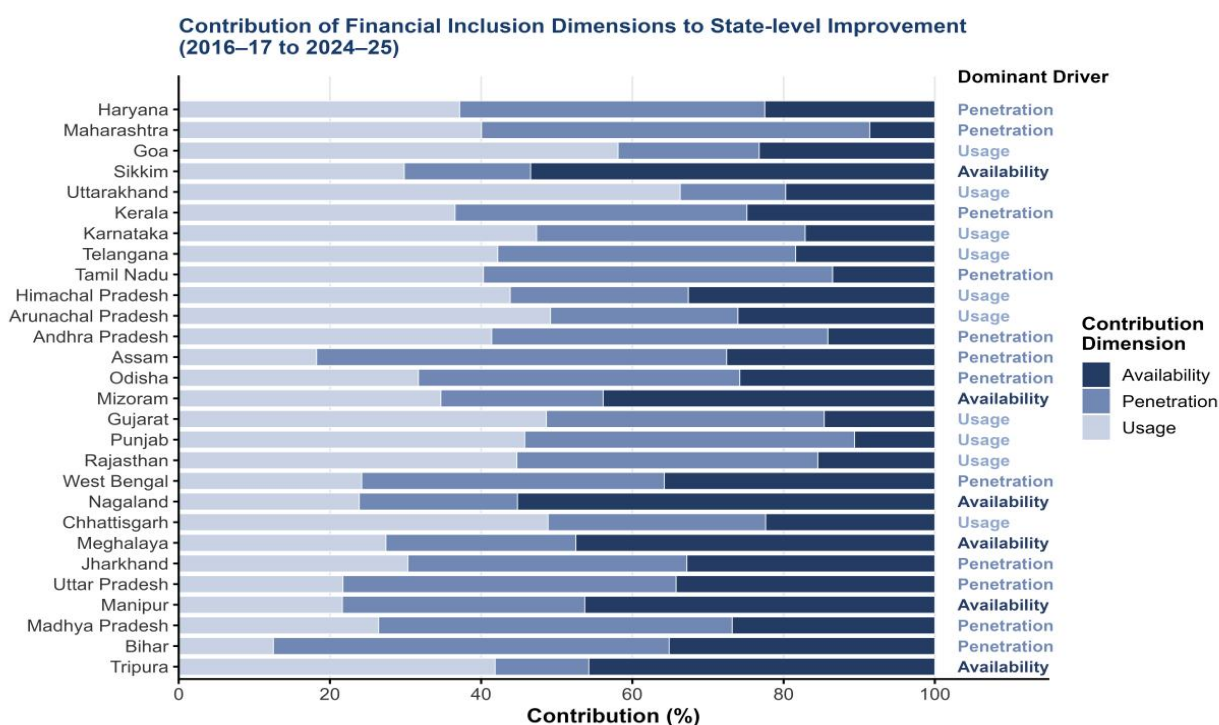
A second group, comprising Maharashtra, Tamil Nadu, Andhra Pradesh, Odisha, Uttar Pradesh, Bihar, Jharkhand, West Bengal and Assam, recorded Penetration as the principal driver of improvement. The presence of both high-performing states such as Maharashtra and lower-ranked states such as Bihar and Uttar Pradesh indicates that Penetration-led growth remains an important pathway across different stages of financial inclusion.

Availability-led growth was concentrated largely in the North-Eastern region. Sikkim, Mizoram, Nagaland, Meghalaya, Manipur and Tripura all recorded Availability as the dominant driver, with only Assam and Arunachal Pradesh as exceptions. This suggests that progress across much of the region continues to depend more on the expansion of financial infrastructure than on deeper penetration or usage.

The decomposition also explains why some states lost rank despite improving their Financial Inclusion Index. Punjab and Himachal Pradesh remained Usage-led but were overtaken by faster-growing states. Tripura, although Availability-led, recorded the smallest increase in the index, while Madhya Pradesh and West Bengal experienced comparatively modest Penetration-led growth. As discussed in Section 6.2.2, these declines reflect slower progress relative to other states rather than deterioration in financial inclusion.

Taken together, the analysis reveals diverse pathways of financial inclusion across Indian states. Usage dominated many of the leading and fastest-improving states, Penetration was the principal driver for the largest group of states, and Availability remained particularly important in much of the North-Eastern region. These distinct patterns provide a richer understanding of interstate differences and form the basis for the Financial Inclusion Development Typology presented in the following subsection.

Figure 6.9 Percentage Contribution of Dimensions to FII Growth



6.5.3 Financial Inclusion Development Typology (FIDT) and Policy Implications

While the Financial Inclusion Index measures the overall level of financial inclusion, states with similar index values may reach those outcomes through different pathways. Building on the dimensional decomposition in Section 6.5.2, the Financial Inclusion Development Typology (FIDT) classifies states according to (i) their 2024–25 level of financial inclusion, namely Financial Inclusion Leaders, Emerging Financial Inclusion States and Financial Inclusion Priority States, and (ii) their dominant growth driver, Penetration, Availability or Usage. The resulting Financial Inclusion Development Matrix (Table 6.6) identifies both a state's level of financial inclusion and the pathway through which it has progressed.

Table 6.6 Financial Inclusion Development Matrix (2024–25)

Financial Inclusion Level	Usage-driven	Penetration-driven	Availability-driven
Financial Inclusion Leaders	Goa, Uttarakhand	Haryana, Maharashtra, Kerala, Tamil Nadu	Sikkim
Emerging Financial Inclusion States	Karnataka, Telangana, Gujarat, Punjab, Rajasthan, Chhattisgarh, Himachal Pradesh, Arunachal Pradesh	Andhra Pradesh, Odisha, Assam, West Bengal	Mizoram, Tripura

Financial Inclusion Level	Usage-driven	Penetration-driven	Availability-driven
Financial Inclusion Priority States	—	Bihar, Uttar Pradesh, Madhya Pradesh, Jharkhand	Meghalaya, Manipur, Nagaland

Source: Author's calculations based on Stage 2 PCA decomposition and Financial Inclusion Development Typology (FIDT).

Several patterns emerge from the typology. First, all three developmental pathways are represented among the Financial Inclusion Leaders, indicating that high financial inclusion can be achieved through different trajectories. Second, Usage dominates the Emerging group, accounting for eight of the fourteen states, suggesting that greater utilisation of financial services has become the principal pathway towards higher financial inclusion. Third, none of the Financial Inclusion Priority States are Usage-driven. Instead, they are either Penetration- or Availability-driven, indicating that improvements in financial service utilisation generally follow earlier progress in financial outreach and infrastructure.

These patterns are reflected in the average Financial Inclusion Index of the three driver groups. Usage-driven states recorded the highest average FII (6.24), followed by Penetration-driven (4.90) and Availability-driven states (3.74). It suggests that Availability-driven growth is concentrated among relatively less financially developed states, whereas Usage-driven growth is more common among financially advanced states. Penetration occupies an intermediate position and is the only pathway represented across all three financial inclusion groups.

To further refine the typology, the dominant driver was complemented by a Driver Strength classification based on the dominance gap between the largest and second-largest dimensional contributions. Driver Strength captures the extent to which improvements in financial inclusion are concentrated in a single dimension rather than distributed more evenly across Penetration, Availability and Usage. States with Strong Driver Strength exhibit a more pronounced dominance of one pathway, whereas Weak Driver Strength indicates that improvements are more balanced across the three dimensions.

Table 6.7 Distribution of Driver Strength by Dominant Driver

Dominant Driver	Weak	Moderate	Strong	Total
Availability	1	2	3	6
Penetration	4	7	1	12
Usage	2	5	3	10

Source: Author's calculations.

As shown in Table 6.7, Driver Strength differs systematically across the three developmental pathways. Availability-driven states are more frequently characterised by Strong Driver Strength (3 of 6 states), indicating that improvements are concentrated in infrastructure and service availability. In contrast, Penetration-driven states are predominantly associated with Moderate (7 states) or Weak (4 states) Driver Strength, suggesting that gains in account ownership generally occur alongside improvements in Availability and Usage rather than being driven by penetration alone. Usage-driven states exhibit a more balanced distribution across Moderate and Strong Driver Strength, reflecting differences in the extent to which financial service utilisation dominates overall financial inclusion improvements.

The Driver Strength classification complements the Financial Inclusion Development Typology by indicating whether recent progress is concentrated in a single dimension or more evenly distributed across Penetration, Availability and Usage. Consequently, states sharing the same dominant driver may nevertheless display different patterns of financial inclusion development.

The typology also provides a basis for differentiated policy design. Financial Inclusion Priority States should address fundamental gaps in financial outreach or infrastructure, depending on whether Penetration or Availability is the primary driver. Emerging Financial Inclusion States, where Usage is the dominant pathway, are likely to benefit most from policies that promote greater adoption of digital payments, formal credit, insurance and pension products. For Financial Inclusion Leaders, future improvements are likely to depend on deepening the use and quality of financial services. These broader policy implications are discussed further in Chapter 7.

In sum, the Financial Inclusion Development Typology (FIDT) demonstrates that states with similar levels of financial inclusion may follow different developmental pathways. By combining financial inclusion levels, dominant drivers and Driver Strength, the framework provides a more nuanced basis for understanding interstate differences and designing targeted policy interventions.

The preceding sections examined how financial inclusion improved and the pathways through which

this progress occurred. The next section considers whether these improvements reduced interstate disparities.

6.6 Interstate Convergence and Persistence

6.6.1 Sigma Convergence Analysis

Sigma convergence examines whether interstate disparities in financial inclusion narrowed over the study period by analysing changes in the cross-sectional dispersion of the Financial Inclusion Index. Because the average level of financial inclusion increased substantially between 2016–17 and 2024–25, both the standard deviation and the coefficient of variation (CV) are examined to distinguish changes in absolute and relative dispersion.

Table 6.8. Sigma Convergence Measures for the Financial Inclusion Index (2016–17 to 2024–25)

Financial Year	Mean FII	Standard Deviation	Coefficient of Variation
2016–17	3.04	1.65	0.543
2017–18	3.29	1.71	0.519
2018–19	3.59	1.86	0.518
2019–20	3.89	1.87	0.481
2020–21	4.11	1.89	0.460
2021–22	4.32	1.99	0.460
2022–23	4.67	2.17	0.465
2023–24	5.02	2.43	0.484
2024–25	5.13	2.49	0.486

Figure 6.10. Sigma Convergence in Financial Inclusion Across Indian States

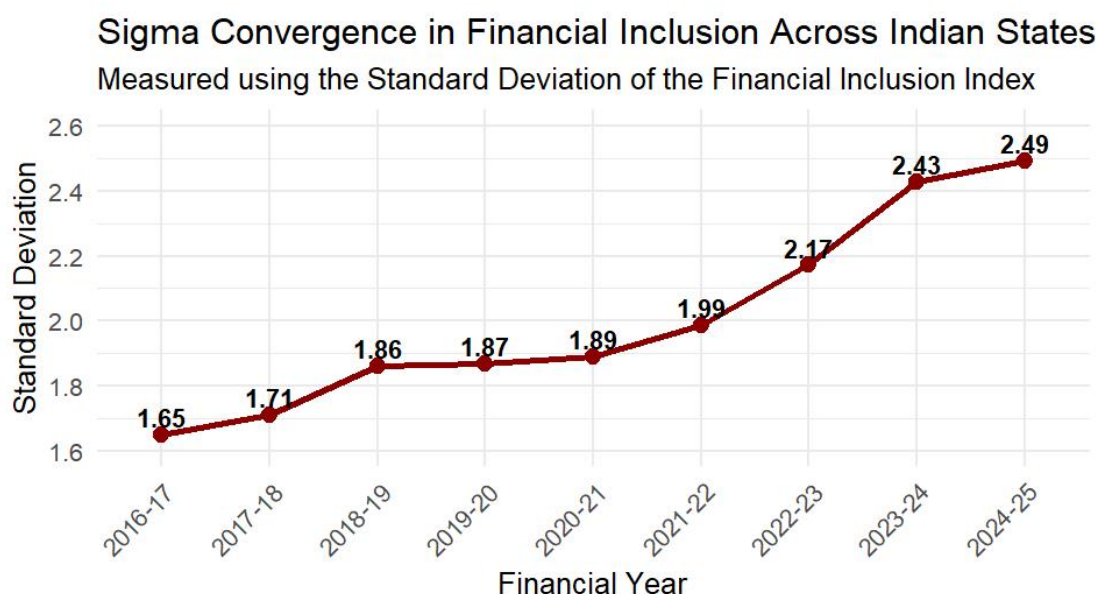


Figure 6.11. Coefficient of Variation of the Financial Inclusion Index

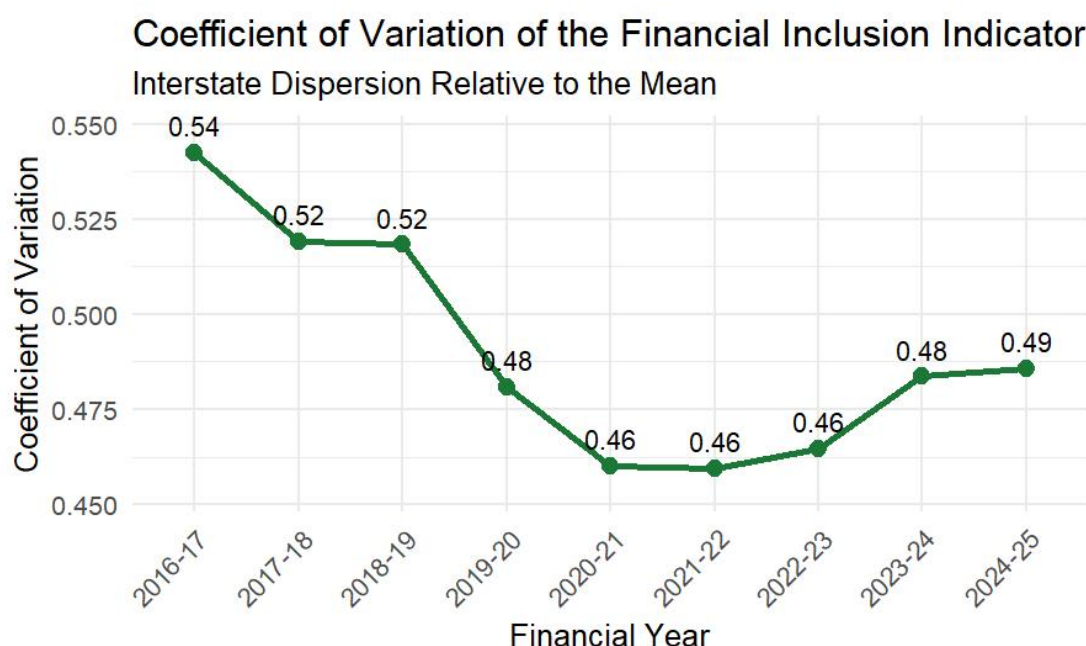


Table 6.8 and Figure 6.10 show that the standard deviation increased steadily from 1.65 in 2016–17 to 2.49 in 2024–25, indicating that absolute interstate disparities widened throughout the study period. Although the increase slowed between 2018–19 and 2020–21, dispersion accelerated after 2021–22, providing no evidence of absolute sigma convergence.

The coefficient of variation presents a different picture. As shown in Figure 6.11, relative dispersion declined from 0.543 in 2016–17 to 0.460 in 2020–21 and remained almost unchanged in 2021–22, indicating a period of relative convergence. It then increased gradually to 0.486 by 2024–25, suggesting a modest reversal. On the whole, the results indicate an initial phase of relative convergence followed by mild relative divergence, although relative dispersion in 2024–25 remained below its 2016–17 level.

Because of Goa's persistently high Financial Inclusion Index, a robustness check recalculated the coefficient of variation after excluding Goa. Although the overall level of dispersion declined, the temporal pattern remained largely unchanged, indicating that the observed trend was not driven solely by Goa's exceptional performance.

Overall, the sigma convergence analysis indicates that improvements in financial inclusion were not accompanied by a sustained reduction in interstate disparities. Absolute disparities widened throughout the study period, while relative convergence proved temporary. These findings motivate

the Markov transition analysis in the following section, which examines whether individual states changed their relative positions within the interstate distribution over time.

6.6.2 Interstate Mobility and Markov Transition Analysis

While the sigma convergence analysis examined changes in overall interstate dispersion, it does not show whether individual states changed their relative position within the financial inclusion hierarchy. To assess the persistence of state-level financial inclusion, a first-order Markov transition analysis classified states into annual quartiles of the Financial Inclusion Index and estimated transition probabilities between consecutive years. The Shorrocks Mobility Index and an alluvial diagram were also used to evaluate overall interstate mobility.

Figure 6.12. Markov Transition Probability Matrix

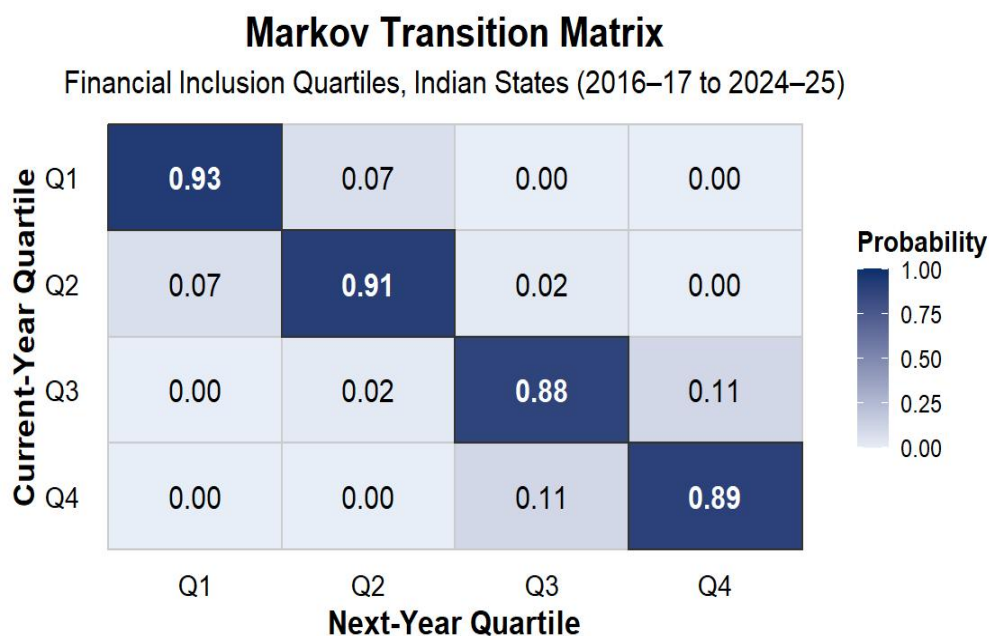
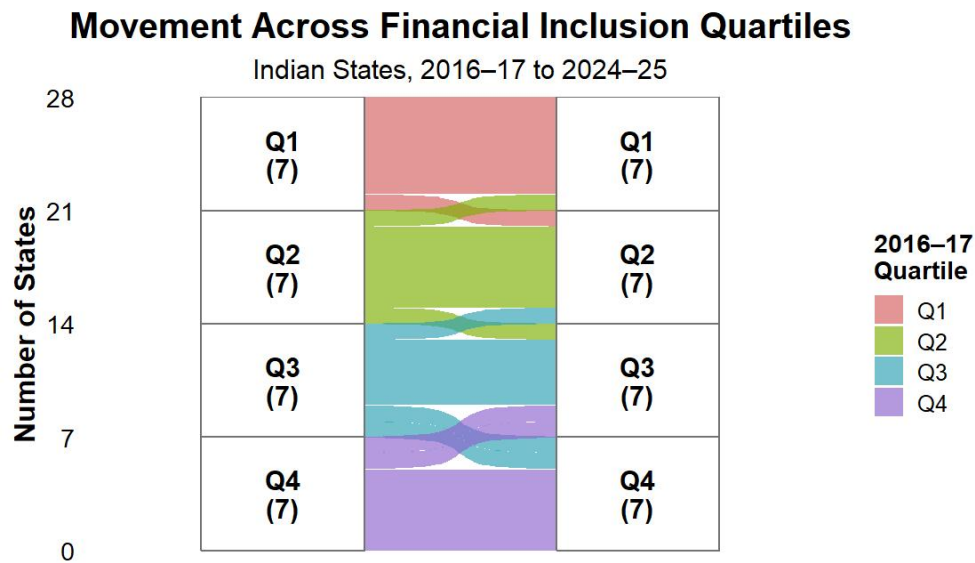


Figure 6.12 indicate a high degree of persistence in interstate financial inclusion. The probability of remaining in the same quartile exceeded 87 per cent across all quartiles, ranging from 87.5 per cent for the third quartile to 92.9 per cent for the lowest quartile. When transitions occurred, they were almost entirely limited to adjacent quartiles. States in the lowest quartile had a 7.1 per cent probability of moving to the second quartile, while those in the highest quartile had a 10.7 per cent probability of moving to the third quartile. Movement between the second and third quartiles was minimal (1.8 per cent in either direction), and no transitions occurred between non-adjacent quartiles. These results indicate that changes in relative financial inclusion were gradual rather than abrupt. The persistence of interstate positions is further supported by the Shorrocks Mobility Index, which recorded a value of 0.131, indicating low interstate mobility over the study period.

Figure 6.13. Interstate Mobility Across Financial Inclusion Quartiles (2016–17 and 2024–25)



The alluvial diagram reinforces these findings by showing that most states remained within their initial financial inclusion quartile between 2016–17 and 2024–25, with only a limited number moving to adjacent quartiles. No state moved across multiple quartiles, confirming the persistence of the interstate financial inclusion hierarchy.

Collectively, the Markov transition probabilities, Shorrocks Mobility Index and alluvial analysis consistently indicate limited interstate mobility in financial inclusion. Together with the sigma convergence results and the high Spearman rank correlation reported in Section 6.2.2, these findings show that improvements in financial inclusion occurred with little reshuffling of states, leaving the overall interstate hierarchy largely unchanged between 2016–17 and 2024–25.

7. DISCUSSION

7.1 Interstate Levels, Regional Disparities and Spatial Patterns of Financial Inclusion

The findings in Chapter 6 show that financial inclusion improved across all Indian states between 2016–17 and 2024–25, but these gains did not substantially reduce interstate disparities. Goa consistently recorded the highest throughout the study period, reflecting the cumulative advantages of a mature financial ecosystem. According to the Goa Economic Survey (2024–25), the state has 795 bank branches operated by 47 banks as of 30 September 2024, with 493 branches (62.01%) located in semi-urban areas and 302 (37.99%) in rural areas among the highest levels of banking accessibility in India. Combined with a service-oriented economy driven by tourism, trade and hospitality, this extensive banking network has supported greater adoption of digital payments,

insurance, credit and other formal financial services. In contrast, Bihar remained the lowest-ranked state despite steady improvements in financial inclusion, reflecting deeper structural constraints: 87.7% rural population, an industry share of just 17.2% of GSVA (vs. 29.3% average), and a Credit-Deposit Ratio of 40.1% against a national 71.7% (NITI Aayog, 2025).

Western India consistently recorded the highest average financial inclusion, followed by Southern India, while Central and Eastern India remained below the national average. This pattern is consistent with previous studies identifying Western and Southern India as India's most financially developed regions (Kasturi & Annemalla, 2024; Yadav et al., 2021). Western India's stronger performance reflects higher income, urbanisation, industrialisation and financial infrastructure, while Southern India has also benefited from institutional innovations such as NABARD's Self-Help Group-Bank Linkage Programme, which has strengthened savings, formal credit access and women's financial participation.

7.2 Financial Inclusion Growth with Persistent Hierarchies

Financial inclusion growth between 2016–17 and 2024–25 was uneven across states even as every state's Financial Inclusion Index rose. This pattern is supported by limited transitions in the Markov transition matrix, a low Shorrocks Mobility Index and the absence of sustained sigma convergence. Together, these findings indicate broad-based financial deepening without interstate convergence.

The pace of improvement differed substantially across states. Haryana, Maharashtra and Gujarat recorded much stronger gains than Tripura and several North-Eastern states, indicating that the benefits of national initiatives depend on state-specific socioeconomic and institutional conditions. This finding is consistent with Kumar (2013) and Ghosh (2012), who identify income, banking infrastructure, literacy, urbanisation, institutional quality, telecommunications, road connectivity and public expenditure as key determinants of interstate financial inclusion.

Relative disparities narrowed between 2016–17 and 2021–22 before reversing into mild relative divergence thereafter, and most transitions remained confined to adjacent quartiles throughout the study period. The timing of this reversal coincides with India's broader post-pandemic "K-shaped" recovery, in which formal-sector credit and digital adoption expanded disproportionately relative to informal and rural channels. The absence of sustained convergence in financial inclusion parallels a broader pattern documented in the Indian regional growth literature, where unconditional divergence in per capita output persists across states unless structural differences in sectoral composition, credit-to-GSDP ratios and population growth are explicitly accounted for (Ghosh & Kaustubh, 2025). This suggests that financial inclusion, like broader economic performance, is unlikely to converge across

states without corresponding convergence in these underlying structural conditions. Together with Erra and Acharya (2021) evidence on structural determinants, these results suggest that financial inclusion progresses gradually and unevenly, with periods of relative convergence vulnerable to reversal when growth becomes concentrated in channels such as digital usage and formal credit that are themselves unevenly distributed across states.

7.3 Spatial Organisation of Financial Inclusion

Global Moran's I and Local Indicators of Spatial Association (LISA) indicate that neighbouring states tend to exhibit similar levels of financial inclusion, suggesting that financial inclusion is shaped by both state-specific characteristics and shared regional conditions. The consistently positive and statistically significant Global Moran's I values between 2016–17 and 2024–25 provide strong evidence of persistent positive spatial autocorrelation. Moran's I increased from 0.339 in 2016–17 to 0.416 in 2022–23 before moderating slightly during the final two years while remaining statistically significant throughout. These findings indicate persistent clustering of financially advanced and financially lagging states, consistent with Erra and Acharya (2021).

The Local Moran's I analysis identifies where these spatial relationships are strongest. In 2024–25, Karnataka was the only High-High cluster, reflecting its strong banking infrastructure, higher literacy, digital adoption and diversified economy, consistent with earlier evidence for Southern India. In contrast, Jharkhand and Assam formed Low-Low clusters, reflecting geographically concentrated financial exclusion associated with weaker infrastructure and socioeconomic development. Similar neighbourhood effects are also reported by Yadav and Reddy (2023).

An equally important finding is the absence of statistically significant High-Low and Low-High spatial outliers. Financially advanced states were generally not adjacent to lagging states, and lagging states were not adjacent to highly developed ones, indicating that spatial clustering in India's financial inclusion landscape is homogeneous. This is consistent with Yadav and Reddy (2023), who report a Global Moran's I of approximately 0.54 for financial inclusion across 591 Indian districts, confirming that spatial dependence operates at multiple administrative scales rather than being an artifact of state-level aggregation.

7.4 Multidimensional Pathways of Financial Inclusion

The decomposition into Penetration, Availability and Usage reveals substantial heterogeneity, with states often reaching similar levels of financial inclusion through different combinations of these dimension. A key finding is the transition from an Availability-led model in 2016–17 to a Usage-led

model by 2024–25. During the early years, financial inclusion was driven primarily by the expansion of banking infrastructure and service availability. By the end of the study period, Usage had become the dominant dimension, indicating that progress increasingly reflected greater utilisation of existing financial services rather than continued expansion of infrastructure. This suggests that financial inclusion in India has entered a more mature phase characterised by financial deepening. This national transition is directly visible in India's own composite FI-Index, which stood at 53.9 in 2021, driven predominantly by the Access sub-index (73.3), before usage growth became the principal contributor to the increases recorded in FY24 (64.2) and FY25 (67.0) (RBI, 2025). The transition also closely aligns with the Global Findex Database 2021, which show that recent progress has increasingly been driven by greater use of existing accounts, particularly through digital payments, rather than continued growth in account ownership. The emergence of Usage as the dominant driver in states such as Goa, Uttarakhand, Karnataka, Telangana and Gujarat is also consistent with Barik, Lenka and Parida (2022), who show that improvements in income, education and human development encourage greater use of formal financial services.

The FIDT further extends this by jointly considering both the overall level of financial inclusion and the dominant growth dimension. Leader states were represented across all three pathways, demonstrating that there is no single route to achieving high financial inclusion. Usage-driven Leaders included Goa and Uttarakhand, Penetration-driven Leaders comprised Haryana, Maharashtra, Kerala and Tamil Nadu, while Sikkim represented an Availability-driven Leader. Most Emerging Financial Inclusion States were Usage-driven, whereas Financial Inclusion Priority States depended primarily on Penetration or Availability, indicating that sustained financial service utilisation generally develops only after adequate outreach and infrastructure have been established. This interpretation is consistent with the World Bank's view that account ownership represents an initial stage of financial inclusion, while sustained usage determines meaningful financial inclusion.

A further contribution is the refinement of the FIDT through the introduction of Driver Strength which demonstrates that dominant pathways differ not only in direction but also in intensity. Availability was more frequently associated with Strong Driver Strength, whereas Penetration generally exhibited Moderate or Weak dominance, suggesting that improvements in account ownership were often accompanied by parallel gains in the other dimensions. This concentration of Strong Availability-driven growth in the North-East is corroborated by an existing real-world policy precedent. The RBI's Payments Infrastructure Development Fund (PIDF), operationalised in January 2021 specifically to subsidise payment acceptance infrastructure in tier-3 to tier-6 centres and the North-Eastern states, raised its subsidy rate for these regions to 90% of device cost from October 2023 (RBI, 2023). The scheme's design explicitly targeting the same states this study identifies as

Availability-Strong and Priority-tier indicates that the RBI had already identified and begun addressing the structural gap this typology captures. These findings carry direct implications for policy design, addressed further in Chapter 8.

8. POLICY IMPLICATIONS AND RECOMMENDATIONS

8.1 National Policy Priorities

The findings of this study indicate that financial inclusion policy in India has entered a new phase. During the early years of the study period, Availability was the strongest contributor to financial inclusion, reflecting the importance of expanding banking infrastructure and financial outreach. By 2024–25, however, Usage had become the dominant contributor at the national level (Figure 6.10), indicating a shift from expanding access to encouraging active participation in the formal financial system. Future policy should therefore place greater emphasis on increasing the use of savings, credit, insurance, pension and digital payment services, while maintaining access where necessary.

This transition has not occurred uniformly across states. As discussed in Section 6.5.2, Availability remains the dominant driver in several geographically remote and infrastructurally constrained states, particularly in the North-East, highlighting continuing gaps in banking outreach, digital connectivity and institutional access. Financial inclusion policy should therefore adopt a differentiated national strategy. Infrastructure expansion should remain the priority in underserved states, while financially advanced states should focus on financial deepening, product diversification and sustained financial service usage.

8.2 Differentiated Policy Framework Based on the Financial Inclusion Development Typology (FIDT)

The Financial Inclusion Development Typology (FIDT) developed in this study (Table 6.6) shows that states with similar levels of financial inclusion often achieve these outcomes through different combinations of Penetration, Availability and Usage. This demonstrates that a uniform policy approach is unlikely to be equally effective across states. Instead, interventions should reflect both the level of financial inclusion and the pathway through which it has been achieved.

For Financial Inclusion Leaders, policy should focus on sustaining progress through financial innovation, advanced digital financial services, greater insurance and pension coverage, improved financial products and better service quality. As these states already have relatively mature financial systems, future gains are likely to come from financial deepening rather than expanding basic

banking infrastructure.

For Emerging Financial Inclusion States, where Usage is the dominant pathway for most states, policy should focus on converting existing financial access into sustained financial participation by encouraging regular account usage, expanding productive credit, increasing insurance and pension enrolment, and promoting digital payments beyond urban centres.

In contrast, Financial Inclusion Priority States consist entirely of Penetration and Availability driven states, with no Usage driven state identified (Table 6.6). Their immediate priority should be expanding banking outreach, strengthening Business Correspondent networks, improving digital connectivity and extending financial infrastructure. Greater financial service usage is likely to follow only after these basic constraints are addressed.

The FIDT offers an evidence-based framework for designing differentiated financial inclusion policies by identifying both the level of financial inclusion achieved and the pathway through which progress has occurred.

8.3 Calibrating Policy Through Driver Strength

A key methodological contribution of this study is the incorporation of Driver Strength, which classifies the dominant dimension as Strong, Moderate or Weak (Table 6.7). This extends the FIDT by recognising that states with the same dominant pathway may require different policy responses depending on the intensity of that dominance.

The findings show that Availability has the highest concentration of Strong Driver states, indicating that infrastructure constraints remain particularly severe and require continued investment in banking outreach, digital connectivity and institutional presence. Rather than proposing a generic infrastructure recommendation, this finding points to a specific, already-tested instrument. The RBI's Payments Infrastructure Development Fund (PIDF), operationalised in January 2021 to subsidise payment acceptance infrastructure specifically in the North-Eastern states, tier-3 to tier-6 centres and the Union Territories of Jammu & Kashmir and Ladakh, raised its subsidy rate for these regions to 90% of device cost from October 2023 (RBI, 2023). The scheme ran until its expiry on 31 December 2025, after which it was allowed to lapse without extension or replacement (Business Standard, 2026). Given this study's finding that Meghalaya, Manipur and Nagaland remain both Availability-Strong and Financial Inclusion Priority states, this indicates that the mechanism best suited to closing their specific gap already existed, was demonstrably targeted at the correct states, and was withdrawn rather than scaled up. This study therefore recommends reinstating or extending

a PIDF-equivalent instrument specifically for states classified as Availability-Strong and Priority-tier under the FIDT, rather than introducing a new scheme from first principles.

In contrast, Penetration is largely characterised by Moderate and Weak dominance, suggesting that improvements in account ownership are generally accompanied by progress in Availability and Usage. Usage driven states are distributed across all three strength categories, reflecting different stages of financial deepening.

Accordingly, Strong Driver states should strengthen weaker dimensions while maintaining their existing advantage. Moderate Driver states require balanced interventions with greater emphasis on lagging dimensions, whereas Weak Driver states are more likely to benefit from integrated, multidimensional strategies. Together, the FIDT and Driver Strength provide a comprehensive policy framework that identifies both the dominant pathway and its intensity, allowing interventions to be tailored more effectively across states.

8.4 Institutional and Regional Policy Implications

The convergence analysis shows that although financial inclusion improved across all states, interstate disparities remained persistent. The increase in the standard deviation of the Financial Inclusion Index, together with the low Shorrocks Mobility Index (0.131) and the high persistence observed in the Markov transition analysis (Section 6.6), indicates that states largely remained within their existing levels of financial inclusion. These findings suggest that broad national programmes alone are unlikely to reduce regional disparities and that sustained support for relatively less financially developed states remains necessary.

The regional and spatial analyses further show that financial inclusion follows clear geographical patterns. A statistically significant Low-Low cluster in eastern and north-eastern India and a High-High cluster centred on Karnataka indicate that neighbouring states often share similar structural conditions. This pattern supports greater regional coordination in expanding financial infrastructure, digital connectivity and institutional outreach. At the same time, the Central Region records the highest within-region coefficient of variation, demonstrating that regional averages can conceal substantial internal disparities. Regional initiatives should therefore be complemented by state and district specific interventions.

Finally, the strong agreement between the proposed Two-Stage PCA Financial Inclusion Index and the Reserve Bank of India's Financial Inclusion Index suggests that the methodology developed in this study provides a reliable framework for state-level monitoring. The RBI may therefore consider

publishing an official state-level Financial Inclusion Index together with separate Penetration, Availability and Usage scores. Such a multidimensional framework would enable policymakers to identify the source of financial inclusion improvements and design more targeted, evidence-based and differentiated policies across Indian states.

9. CONCLUSION

9.1 Summary of the Study

This study set out to construct and compare state-level Financial Inclusion Indices (FII) for 28 Indian states over the period 2016–17 to 2024–25 using a hierarchical Two-Stage Principal Component Analysis (PCA), and to examine interstate disparities in financial inclusion. Specifically, the study sought to: (i) construct and validate a Two-Stage PCA-based Financial Inclusion Index against the Reserve Bank of India's Financial Inclusion Index (RBI FI-Index) before extending it to the state panel; (ii) construct a parallel state-level Financial Inclusion Index using Sarma's distance-based methodology; (iii) compare the two methodologies to assess their consistency and robustness; and (iv) analyse the spatial and temporal evolution of financial inclusion across Indian states, including regional disparities, convergence dynamics and interstate mobility. These objectives have been addressed through the methodological, comparative and empirical analyses presented in Chapters 4 to 8.

9.2 Summary of Key Findings

The empirical evidence reveals that although financial inclusion improved across all states during 2016–17 to 2024–25, interstate disparities remained persistent. States with relatively high levels of financial inclusion at the beginning of the study largely maintained their advantage throughout the period, while lower-performing states continued to lag behind despite overall progress. The spatial distribution of financial inclusion also exhibited a clear regional pattern, with relatively higher levels concentrated in western and southern India and comparatively lower levels observed across much of eastern, central and parts of north-eastern India. These findings suggest that financial inclusion in India has expanded in a geographically uneven manner rather than converging towards a common level.

Beyond measuring overall levels of financial inclusion, the study demonstrates that similar index values can conceal fundamentally different developmental pathways. Although the national pattern evolved from an Availability-led to a Usage-led model over the study period, states followed diverse trajectories characterised by different combinations of Penetration, Availability and Usage. The

proposed Financial Inclusion Development Typology (FIDT), together with the Driver Strength framework, extends conventional index-based analysis by identifying not only the level of financial inclusion achieved but also the dominant pathway and the extent to which progress is concentrated in a particular dimension.

The proposed Financial Inclusion Development Typology (FIDT), complemented by the Driver Strength framework, therefore extends conventional index-based analysis by distinguishing not only the level of financial inclusion achieved but also the dominant pathway and the extent to which progress is concentrated in a particular dimension. Together, these frameworks provide a more nuanced understanding of interstate financial inclusion and offer a stronger analytical basis for differentiated policy design.

Finally, the study finds limited evidence of interstate convergence in financial inclusion. While all states experienced improvement, relative positions changed only marginally and regional disparities remained largely intact over time. The persistence of established spatial patterns indicates that improvements in financial inclusion have not automatically translated into greater interstate equality. Consequently, the findings suggest that a uniform national approach is unlikely to be sufficient. Instead, policies should be designed to reflect the specific developmental pathways and constraints of individual states, recognising that different combinations of Penetration, Availability and Usage require different policy priorities.

9.3 Limitations of the Study

First, the analysis relies exclusively on officially published secondary data. Although the data were carefully compiled and validated, some state-level inconsistencies required limited preprocessing before index construction. Second, the selection of indicators was constrained by data availability at the state level. While the study aimed to capture the multidimensional nature of financial inclusion, several theoretically relevant variables were either unavailable or not reported consistently across all states and years. Consequently, in a few cases, proxy indicators were used to represent broader dimensions of financial inclusion. Third, while the selected indicators adequately represent the three dimensions of financial inclusion, they do not capture important aspects such as financial literacy, service quality, affordability, consumer protection and informal financial activity because comparable state-level data are not currently available. Fourth, the analysis is conducted at the state level and therefore does not capture substantial intra-state disparities, particularly between rural and urban areas or across districts.

Finally, the study focuses primarily on measuring and comparing financial inclusion rather than

identifying its underlying determinants or establishing causal relationships. Examining the socioeconomic, institutional and policy factors driving interstate differences remains an important avenue for future research.

9.4 Concluding Remarks

This study has shown that India's progress in financial inclusion during 2016–17 to 2024–25 has been substantial but uneven. Every state recorded improvement, yet longstanding interstate disparities, regional inequalities and spatial clustering remained largely intact. At the same time, the transition from an Availability-led to a Usage-led national profile demonstrates that financial inclusion in India is evolving beyond expanding access towards encouraging meaningful participation in the formal financial system.

More importantly, a single composite score is only a starting point for understanding interstate financial inclusion, not an endpoint: states with similar index values frequently follow different developmental pathways, differing in both direction and intensity. By integrating the Financial Inclusion Development Typology (FIDT) with Driver Strength, this framework strengthens both the measurement and interpretation of financial inclusion across Indian states.

Overall, the validated Two-Stage PCA methodology, together with the multidimensional analytical framework developed in this study, contributes to a deeper understanding of interstate financial inclusion and provides an evidence-based foundation for more differentiated and effective policymaking. As India continues to advance its financial inclusion agenda, recognising the diversity of state-level development pathways will be essential for achieving balanced and inclusive financial development.

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